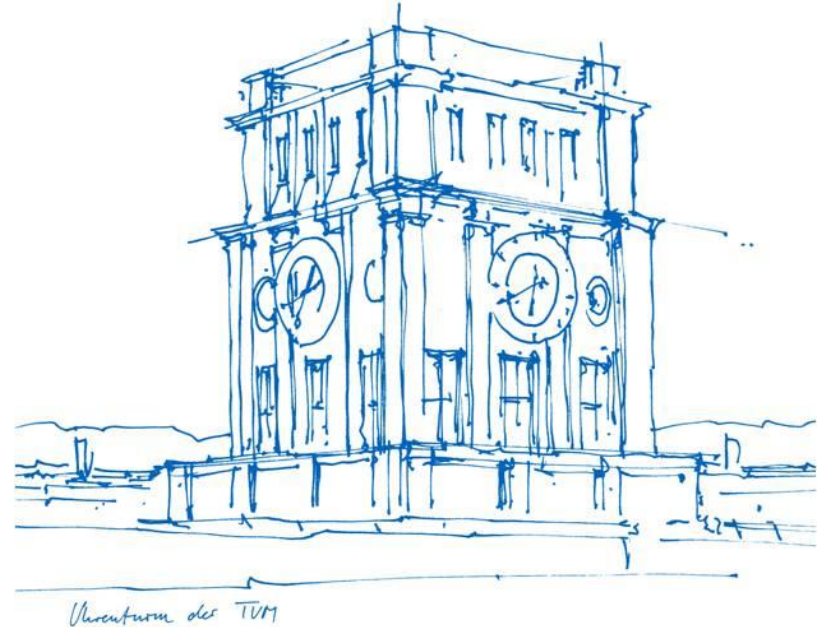


Tutorial - Medical Image Registration

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Huaqi Qiu
Imperial College London

Vasiliki Sideri-Lampretsa
Technical University of Munich

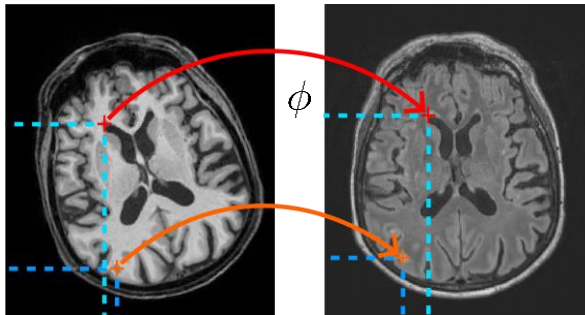


Introduction to Medical Image Registration

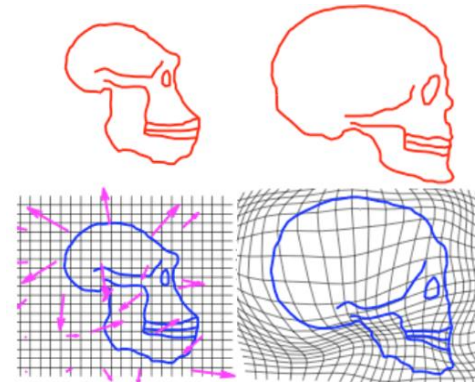
What is medical image registration?

- The aim is to establish spatial correspondences between two or multiple images.
- Image registration is the process of transforming sets of image data into one common coordinate system.

Establish anatomical correspondences



Deform underlying spatial domain

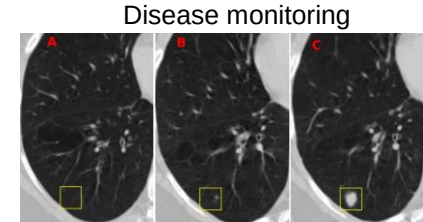


Durrleman et al., *Morphometry of anatomical shape complexes with dense deformations and sparse parameters*, NeuroImage 2014.

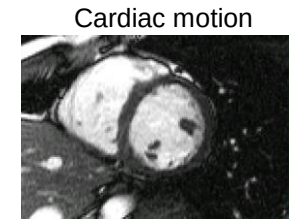
Clinical applications: When do we need it?

Finding **geometric**, **anatomical** and **functional** alignment for

- Disease monitoring
- Motion analysis
- Growth analysis
- Assisted/Guided surgery
- Intervention and treatment planning
- Computer-aided diagnosis
- Population analysis



Tumour growth observed over 2 years from baseline



Gholipour et al., 2017.

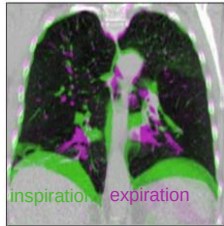


Wright et al., 2022.

Intra-patient image registration

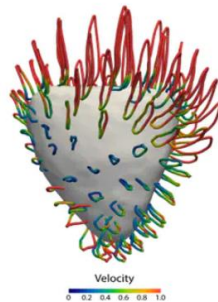
- (Multi-modal) image fusion
- Quantification of change

Quantification of change

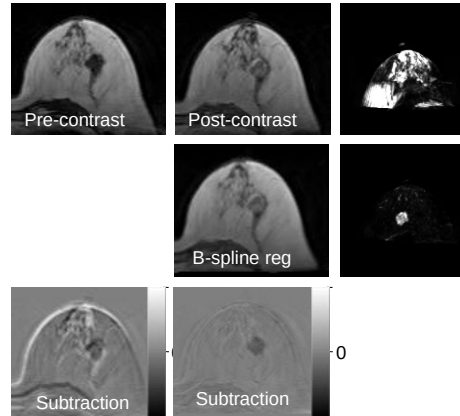


Heinrich, et al., 2013.

Cardiac motion analysis

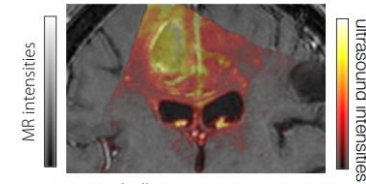


Dynamic image registration



Estimation of respiratory motion

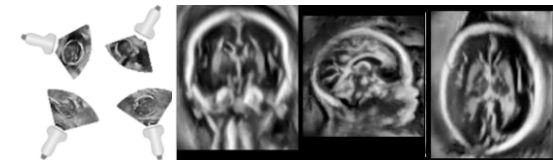
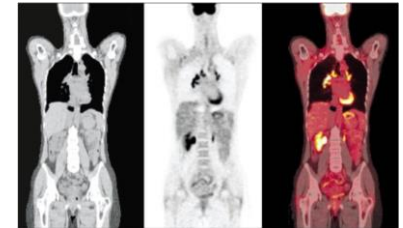
Image fusion



CuRIOUS challenge, 2019

US-guided brain tumour surgery

Multimodal diagnostic by combining CT/PET scans



Wright, et al., 2022.

4D US image compounding

Inter-patient image registration

- Population analysis
- Segmentation

Segmentation

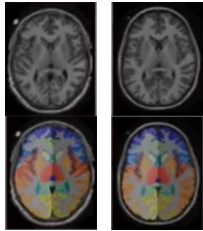


Image
registration

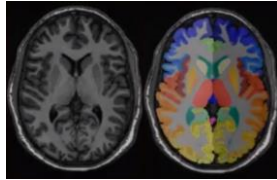


Image
registration



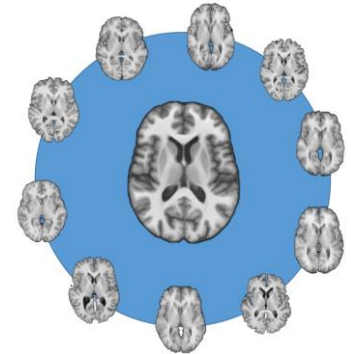
Atlas with expert
segmentations

New patient
scan

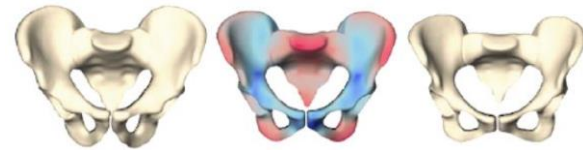
Population analysis



Wright, et al., 2019.



Template / atlas construction



Ambellan, et al., 2019.

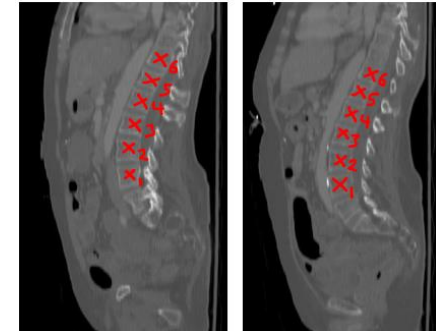
Shape modelling

Introduction to medical image registration

Two main approaches

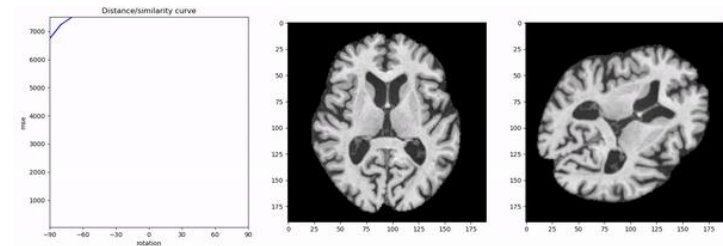
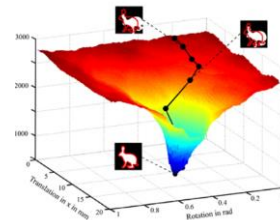
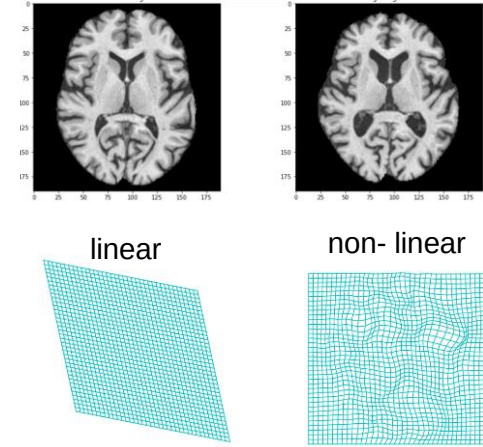
- **Feature-based matching**
 - Identify corresponding structures
 - Fit spatial transformation
- **Intensity-based matching**
 - Define notion of similarity between images
 - Find a spatial transformation which maximizes the similarity

Feature-based registration



Main components

- Fixed image F defined on domain Ω_F
- Moving image M defined on domain Ω_M
- Spatial transformation ϕ mapping from Ω_F to Ω_M
- Transformation model Φ (space of transformations ϕ)
- Interpolation method
- Objective / cost / energy function $\mathcal{J}(\cdot, \cdot, \cdot)$
- Image similarity / distance measure $\mathcal{D}(\cdot, \cdot)$
- Regularization term $\mathcal{R}(\cdot)$
- Optimization method



Generic pairwise image registration algorithm

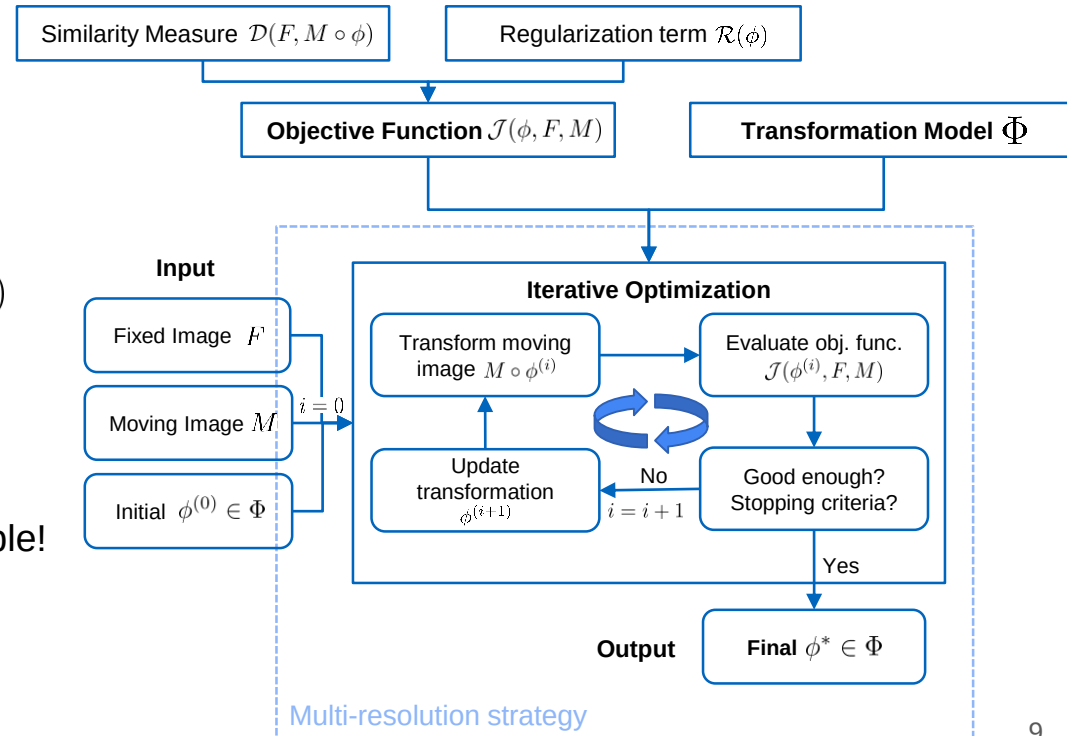
- Optimization problem

$$\phi^* = \underset{\phi}{\operatorname{argmin}} \mathcal{J}(\phi, F, M)$$

- Objective function

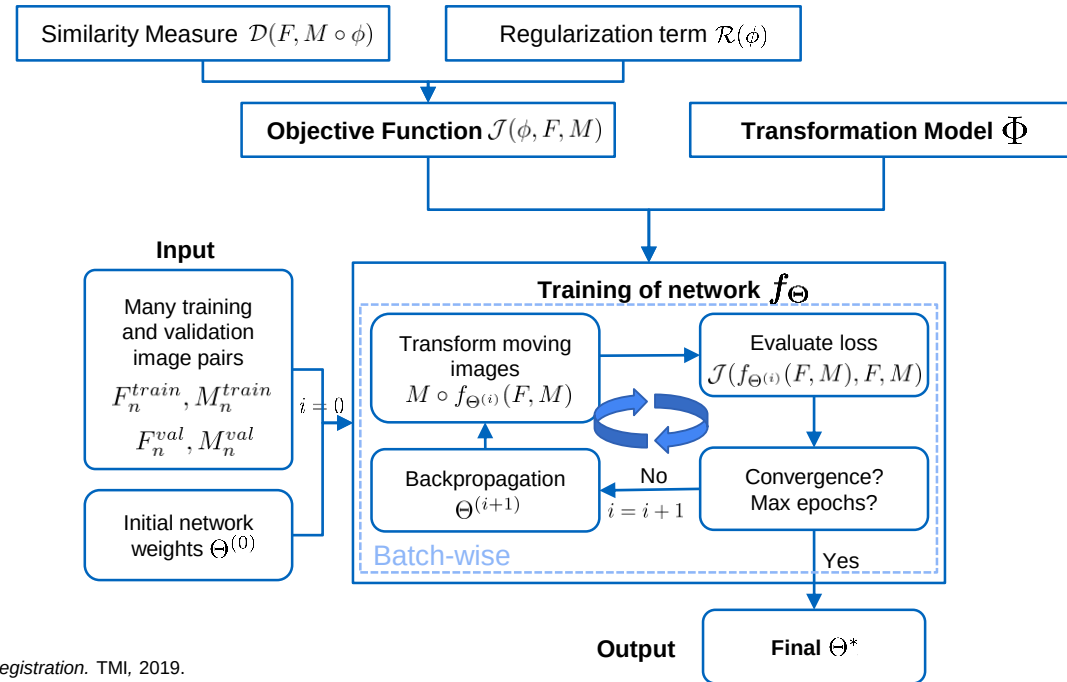
$$\mathcal{J}(\phi, F, M) = \mathcal{D}(F, M \circ \phi) + \alpha \mathcal{R}(\phi)$$

- Solved iteratively
- Often Gradient-based solvers
→ everything needs to be differentiable!



Learning-based image registration algorithm

- Learn the **spatial transformation**
- Exploit information of many image pairs
- Optimization of many training samples instead of over one image pair
- CNN f_{Θ} parametrised by Θ to encode transformation
- Differentiable image sampling
Jaderberg, et al., 2015.
- **Supervised**
- **Unsupervised**
- **Weakly supervised**

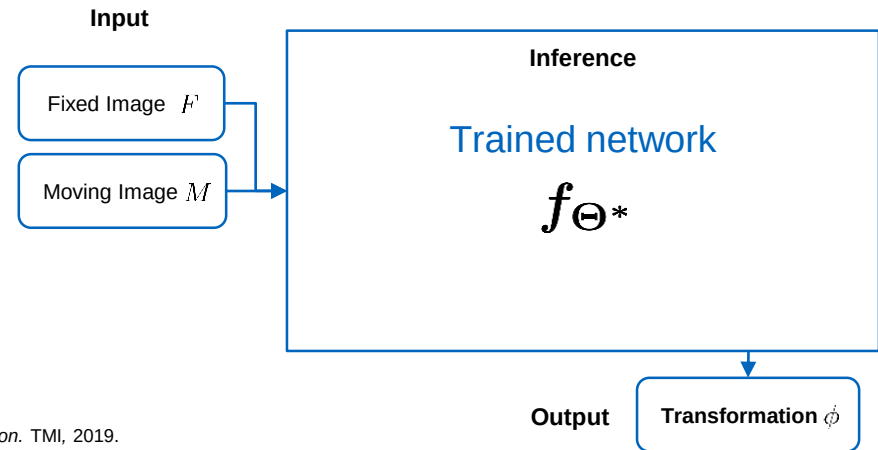


Balakrishnan, et al., *VoxelMorph: a learning framework for deformable medical image registration*. TMI, 2019.
De Vos, et al., *A deep learning framework for unsupervised affine and deformable image registration*. MedIA, 2019.

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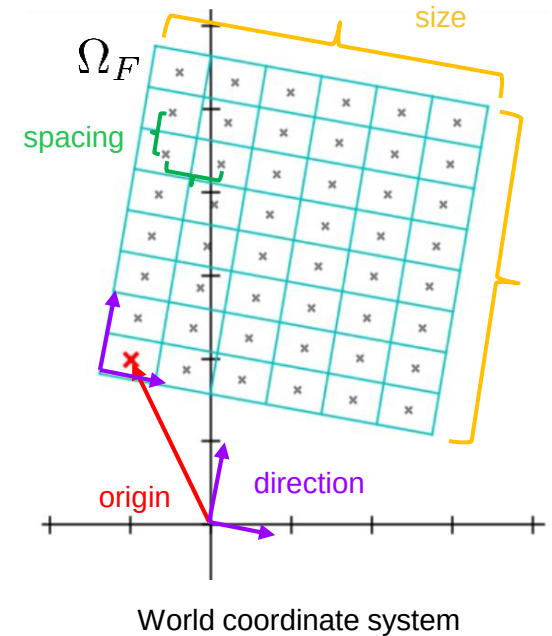
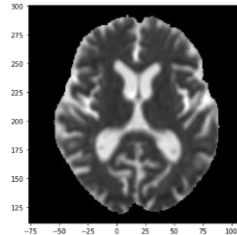
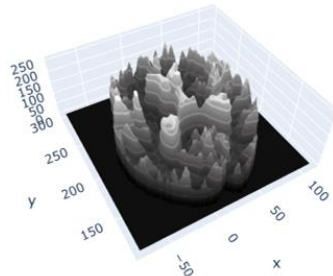


Transformation estimation in a single forward pass

Image Geometry and Image Warping

Medical image representations

- **Continuous:** functions $F, M : \mathbb{R}^d \rightarrow \mathbb{R}$
 $F(\mathbf{x}) \in \mathbb{R}, \mathbf{x} \in \Omega_F \subset \mathbb{R}^d$
- **Discrete:** d-dim. Matrix $F, M \in \mathbb{R}^{m_1 \times \dots \times m_d}$
- **Meta information:**
 - Pixel / voxel spacing
 - Image origin
 - Image direction / orientation



Spatial transformations: Global

- Spatial transformations $\phi : \Omega_F \subset \mathbb{R}^d \rightarrow \Omega_M \subset \mathbb{R}^d$
 $M \circ \phi = M(\phi)$
- **Global (or linear) transformations models (parametric)**

$$\phi : \mathbb{R}^p \times \mathbb{R}^d \rightarrow \mathbb{R}^d$$

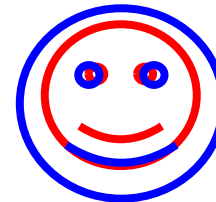
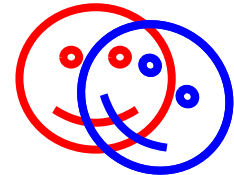
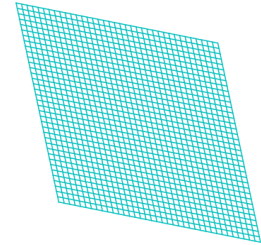
(in homogenous coordinates) $\phi(\theta, \mathbf{x}) = A \cdot \mathbf{x} \quad A = \begin{pmatrix} A_{11} & A_{12} & b_1 \\ A_{21} & A_{22} & b_2 \\ 0 & 0 & 1 \end{pmatrix}$

Affine parameters in 2D:

$$\theta = (A_{11}, A_{12}, b_1, A_{21}, A_{22}, b_2)^T \in \mathbb{R}^6, (d = 2, p = 6)$$

- Rigid: Translation + Rotation
- Similarity: Rigid + Scaling
- Affine: Similarity + Shearing

linear



Spatial transformations: Local

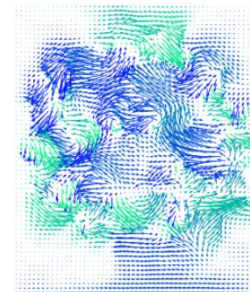
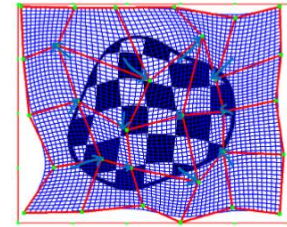
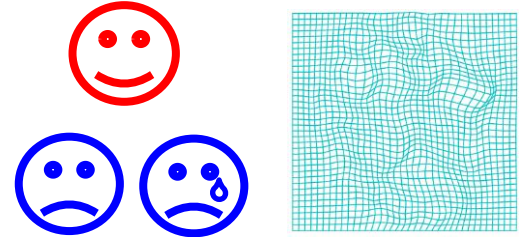
- Spatial transformations $\phi : \Omega_F \subset \mathbb{R}^d \rightarrow \Omega_M \subset \mathbb{R}^d$
 $M \circ \phi = M(\phi)$
- **Local (or non-linear) transformation models**
- **Parametric** transformation models $\phi : \mathbb{R}^p \times \mathbb{R}^d \rightarrow \mathbb{R}^d$

Free-Form-Deformations via B-Splines

e.g., Rueckert, et al., 1999; Klein, et al., 2009 (Elastix).

- **Non-parametric** transformation models $\phi : \mathbb{R}^d \rightarrow \mathbb{R}^d$
e.g., Beg, et al., 2004 (LDDMM); Avants, et al., 2008 (ANTs);
Vercauteren, et al., 2009 (DiffDemons).
 $\phi(\mathbf{x}) = \mathbf{x} + \mathbf{u}$
 - With displacements $\mathbf{u} \in \mathbb{R}^d$
 - Dense displacement fields
 - The transformation model **is** the deformation field

non- linear

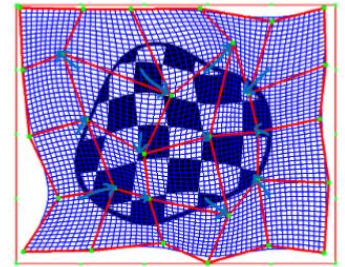


Spatial transformations: Local

Parametric

- **Advantage:**
 - Reduced degrees of freedom (e.g. control points at every 10th pixel only)
 - Reduced memory / storage requirements, faster optimisation
 - May have built-in regularisation to produce smooth deformations
- **Disadvantage:**
 - Loss of local flexibility of movement, due to capture range of parametric representation

$$\phi : \mathbb{R}^p \times \mathbb{R}^d \rightarrow \mathbb{R}^d$$



Non-parametric

- **Advantage:**
 - Full flexibility of “moving” pixels
- **Disadvantage:**
 - Large number of degrees of freedom: number of pixels x image dimension
 - Large memory / storage requirements, slow optimisation
 - Ill-posed: Danger of pixel intersection / collapse, regularization needed!

$$\phi : \mathbb{R}^d \rightarrow \mathbb{R}^d$$

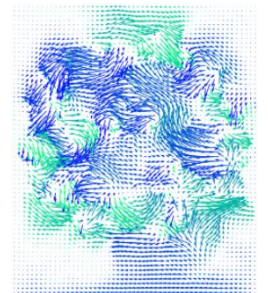
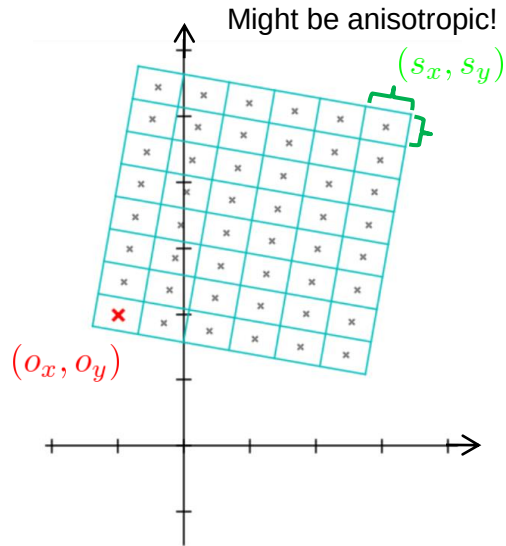


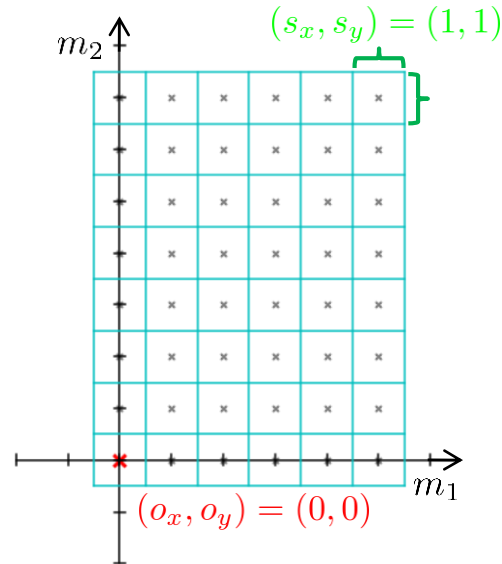
Image coordinate systems

World coordinate system



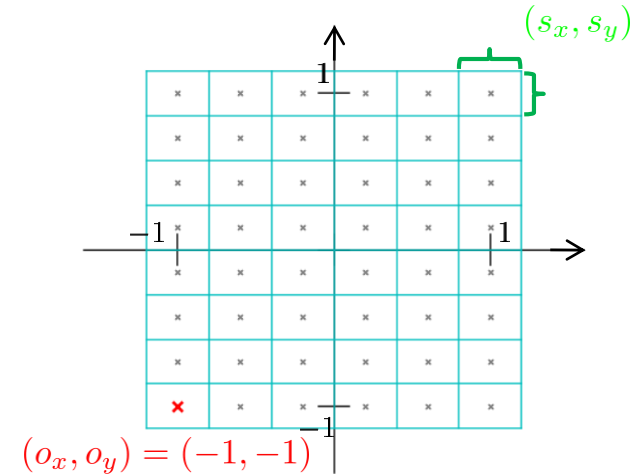
- Origin : $(0,0)$
- Spacing in mm
- Orientation
- Domain Ω

Image coordinate system



- Origin : $(0,0)$
- Spacing in px
- Domain $\Omega = [0, m_1] \times [0, m_2]$

Normalized coordinate system



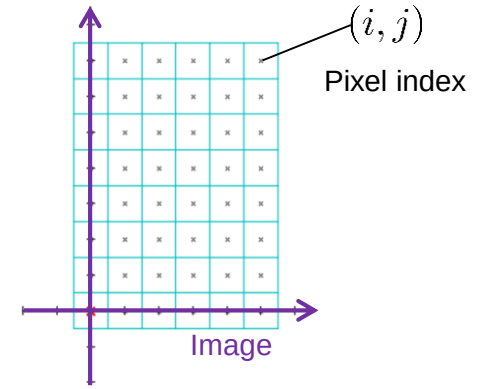
- Origin : $(-1, -1)$
- Spacing : $2 / (sz - 1)$
- Domain $\Omega = [-1, 1] \times [-1, 1]$

Change between coordinate systems

Image to world

$$\begin{pmatrix} x \\ y \\ 1 \end{pmatrix} = T_{i2w} \begin{pmatrix} i \\ j \\ 1 \end{pmatrix}$$

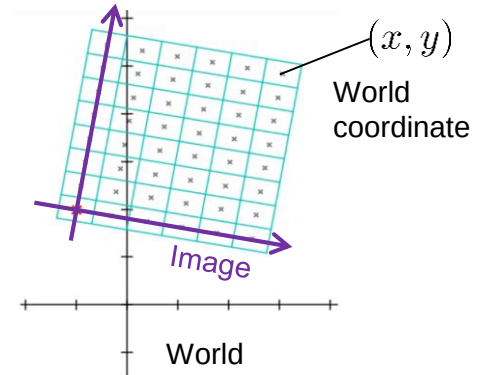
$$T_{i2w} = \begin{pmatrix} 1 & 0 & o_x \\ 0 & 1 & o_y \\ 0 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} d_{xx} & d_{yx} & 1 \\ d_{xy} & d_{yy} & 1 \\ 0 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & 1 \end{pmatrix}$$



World to image

$$T_{w2i} = T_{i2w}^{-1}$$

$$\begin{pmatrix} i \\ j \\ 1 \end{pmatrix} = T_{w2i} \begin{pmatrix} x \\ y \\ 1 \end{pmatrix}$$

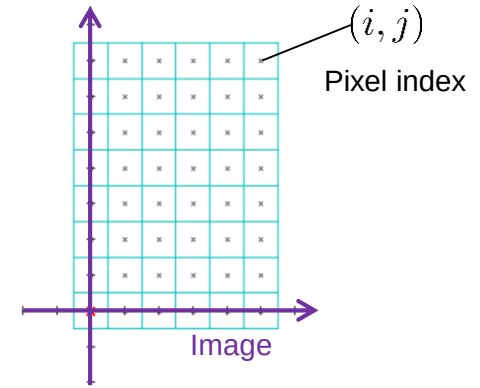


Change between coordinate systems

Image to normalized domain

$$\begin{pmatrix} x \\ y \\ 1 \end{pmatrix} = T_{i2n} \begin{pmatrix} i \\ j \\ 1 \end{pmatrix}$$

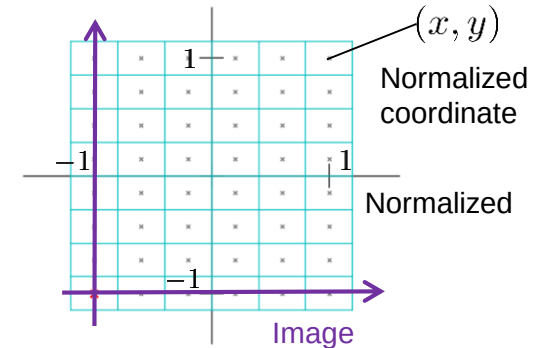
$$T_{i2n} = \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} \frac{2}{m_1-1} & 0 & 0 \\ 0 & \frac{2}{m_2-1} & 0 \\ 0 & 0 & 1 \end{pmatrix}$$



Normalized domain to image

$$T_{n2i} = T_{i2n}^{-1}$$

$$\begin{pmatrix} i \\ j \\ 1 \end{pmatrix} = T_{n2i} \begin{pmatrix} x \\ y \\ 1 \end{pmatrix}$$



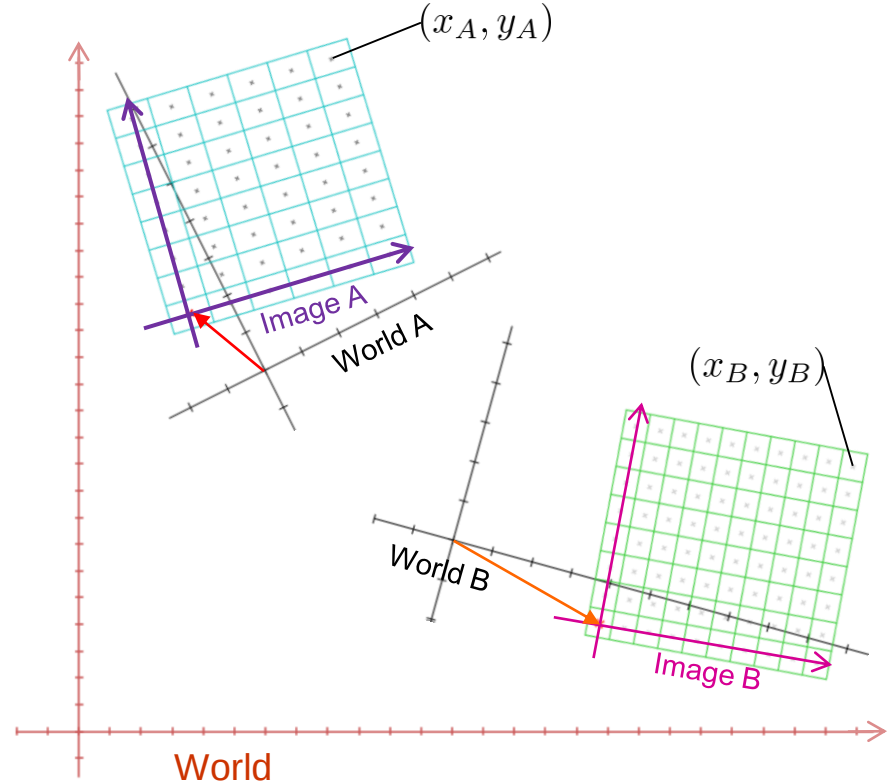
More than one coordinate systems

Image B to image A

$$\begin{pmatrix} x_A \\ y_A \\ 1 \end{pmatrix} = T_{i2w}^A \cdot T_{B2A} \cdot T_{w2i}^B \begin{pmatrix} x_B \\ y_B \\ 1 \end{pmatrix}$$

Initializations:

- Align image centres
- Ignore origin and direction
- Align centres of intensity mass



Spatial transformation in different coordinate systems

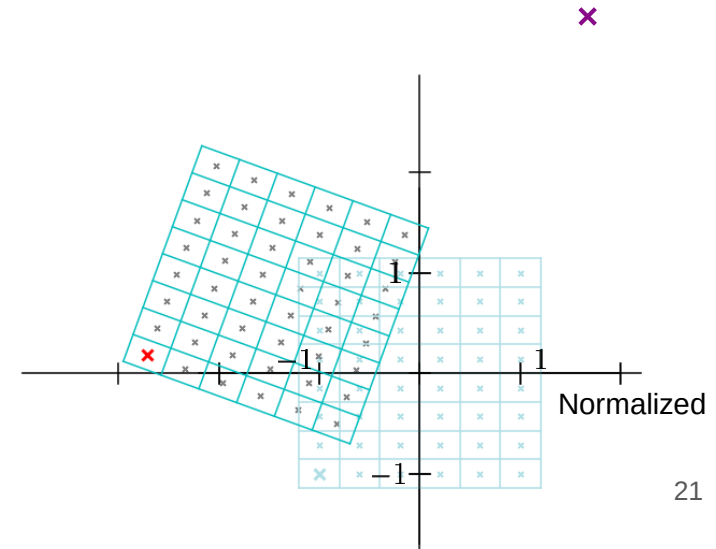
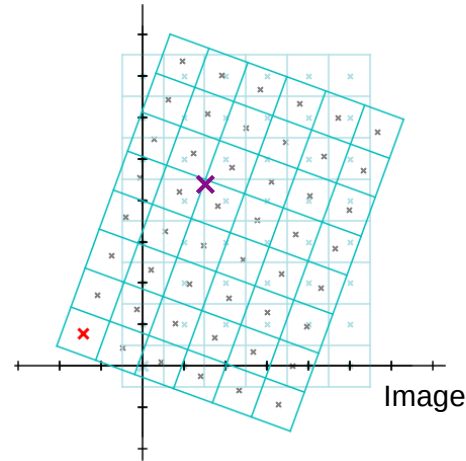
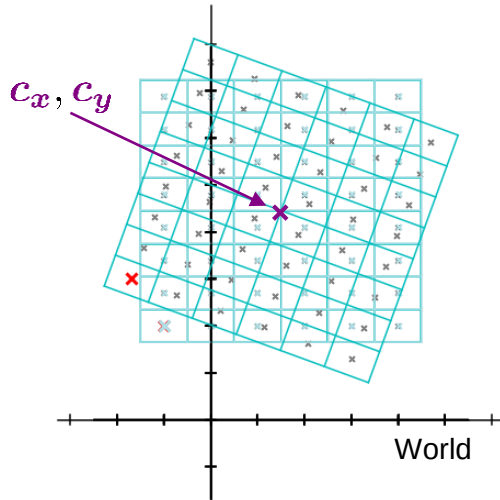
Rigid transformation in world coordinate system

Translations t_x, t_y in mm

Center of rotation c_x, c_y in world coordinates

$$\theta = (\alpha, t_x, t_y)$$

$$\phi(\theta) = \begin{pmatrix} 1 & 0 & c_x \\ 0 & 1 & c_y \\ 0 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} \cos \alpha & -\sin \alpha & t_x \\ \sin \alpha & \cos \alpha & t_y \\ 0 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 1 & 0 & -c_x \\ 0 & 1 & -c_y \\ 0 & 0 & 1 \end{pmatrix}$$



Spatial transformation in different coordinate systems

Rigid transformation in world coordinate system

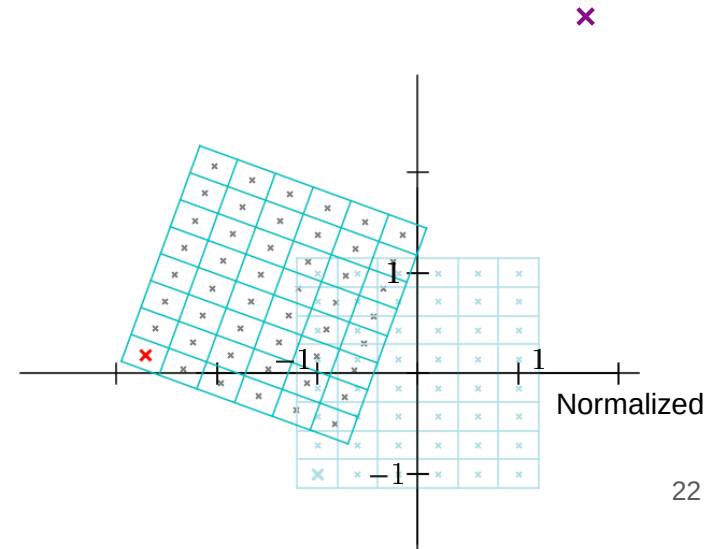
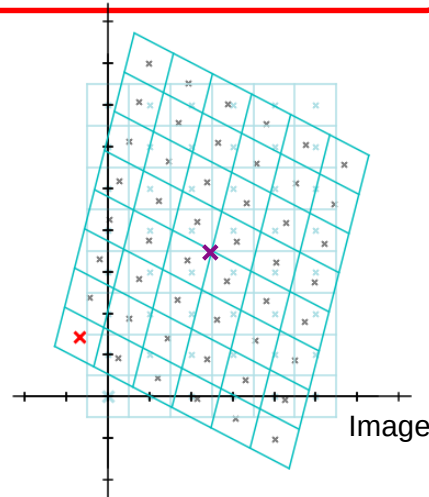
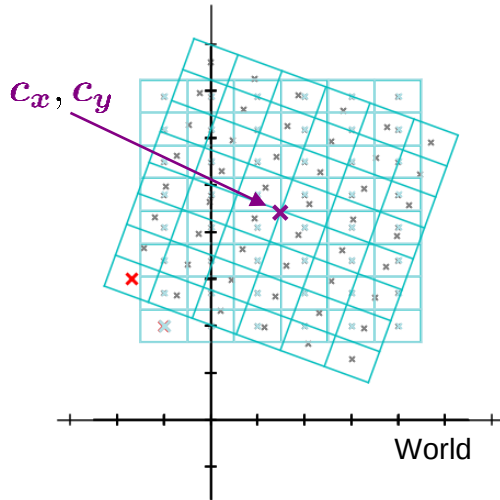
Translations t_x, t_y in mm

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$$\phi_I(\theta) = T_{i2w}^{-1} \cdot \phi(\theta) \cdot T_{i2w}$$



Spatial transformation in different coordinate systems

Rigid transformation in world coordinate system

Translations t_x, t_y in mm

Center of rotation c_x, c_y in world coordinates

$$\theta = (\alpha, t_x, t_y)$$

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$$\phi_I(\theta) = T_{i2w}^{-1} \cdot \phi(\theta) \cdot T_{i2w}$$

$$\begin{aligned} \phi_N(\theta) &= T_{n2w}^{-1} \cdot \phi(\theta) \cdot T_{n2w} \\ T_{n2w} &= T_{i2w} \cdot T_{i2n}^{-1} \end{aligned}$$

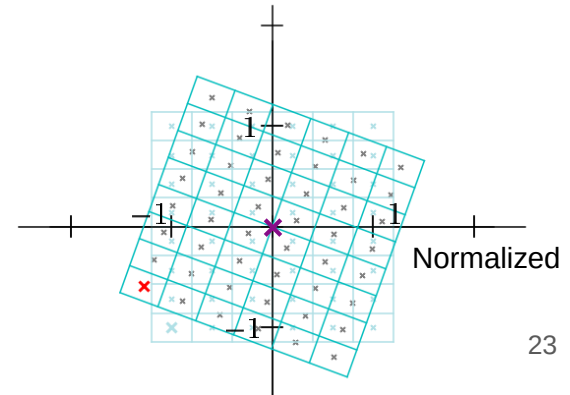
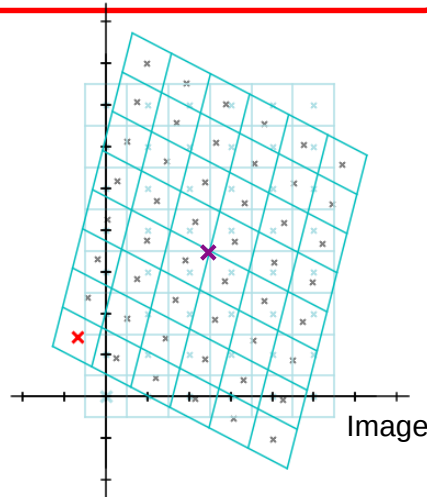
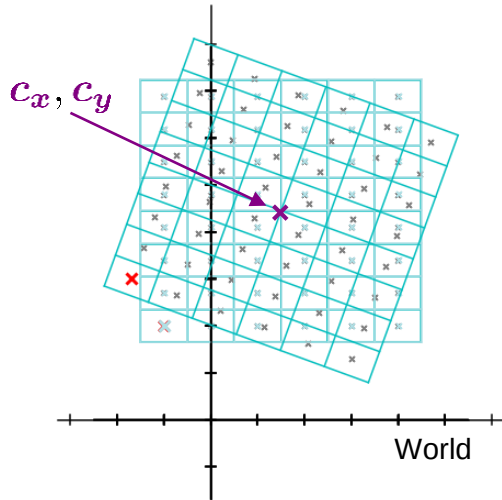


Image interpolation

- Given a spatial transformation $\phi : \Omega_F \rightarrow \Omega_M$ between image M and F . In general, $\phi(\mathbf{x})$ does not fall on a voxel position.
- Every resampling introduces errors. Avoid resampling many times!

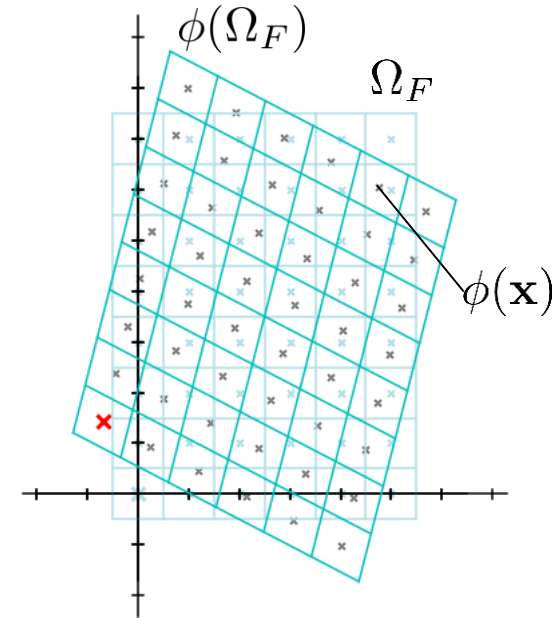
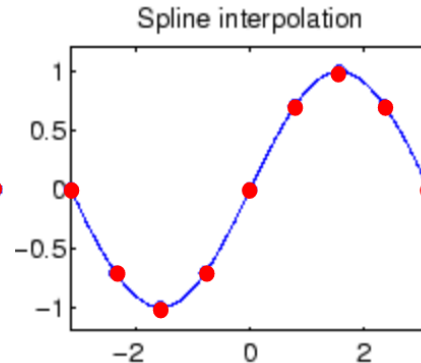
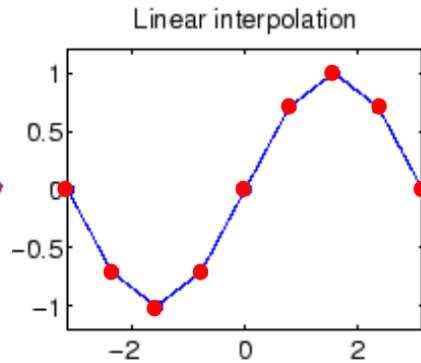
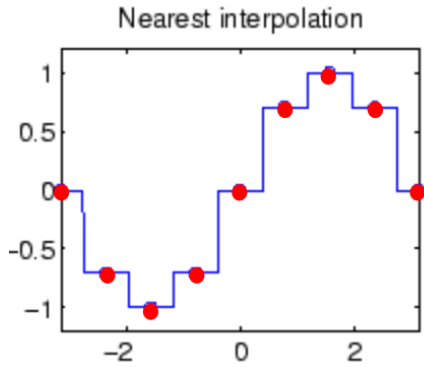
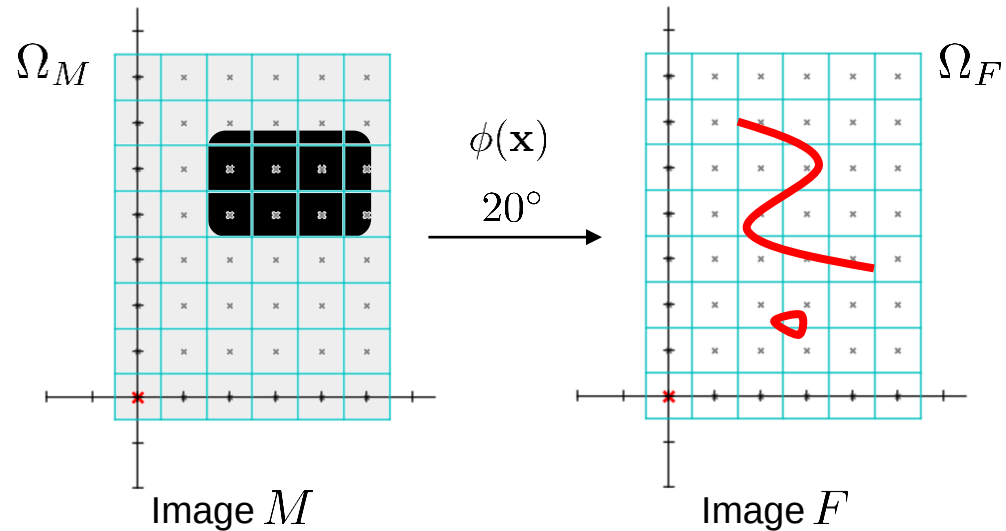


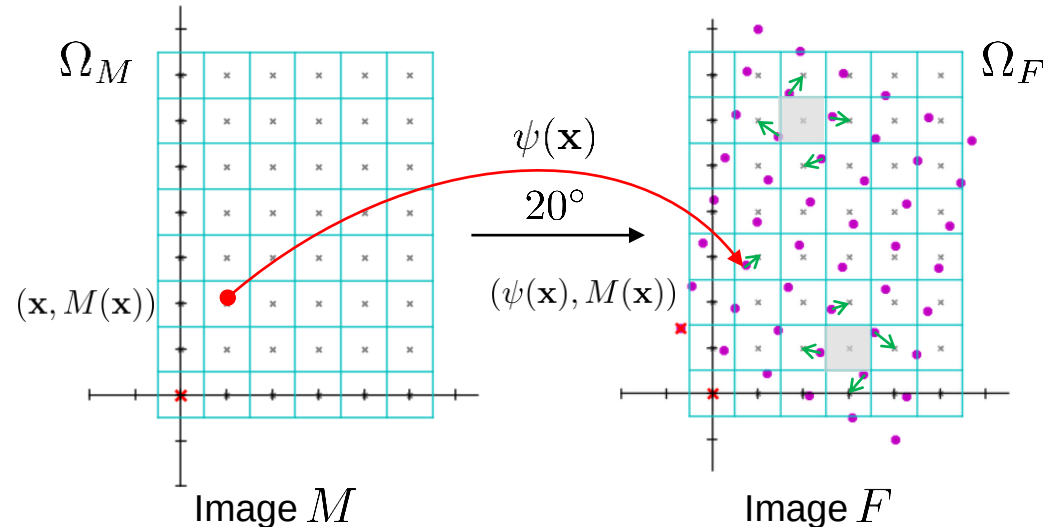
Image warping

- How to apply a transformation to an image?
- Two approaches: **Forward** and **backward mapping**



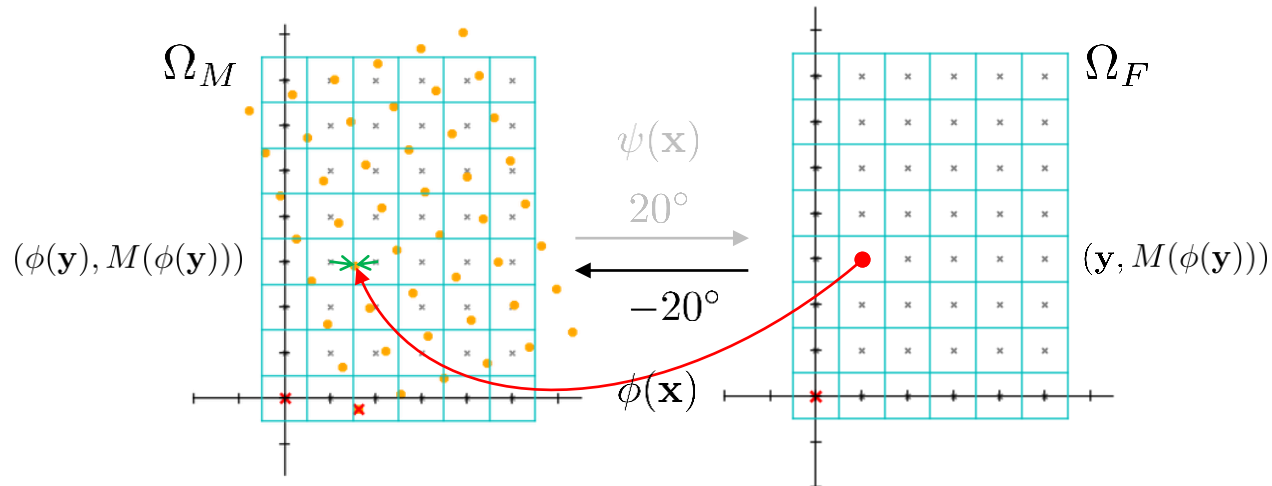
Forward mapping (Lagrange approach)

- Consider a grid point $\mathbf{x} \in \Omega_M$ in image M and a spatial transformation $\psi : \Omega_M \rightarrow \Omega_F$
- The intensity value $M(\mathbf{x})$ at position $\mathbf{x}$ is moved to the new position $\psi(\mathbf{x})$.
- Interpolation to regular grid points.
- Forward mapping might result in holes in the warped image!



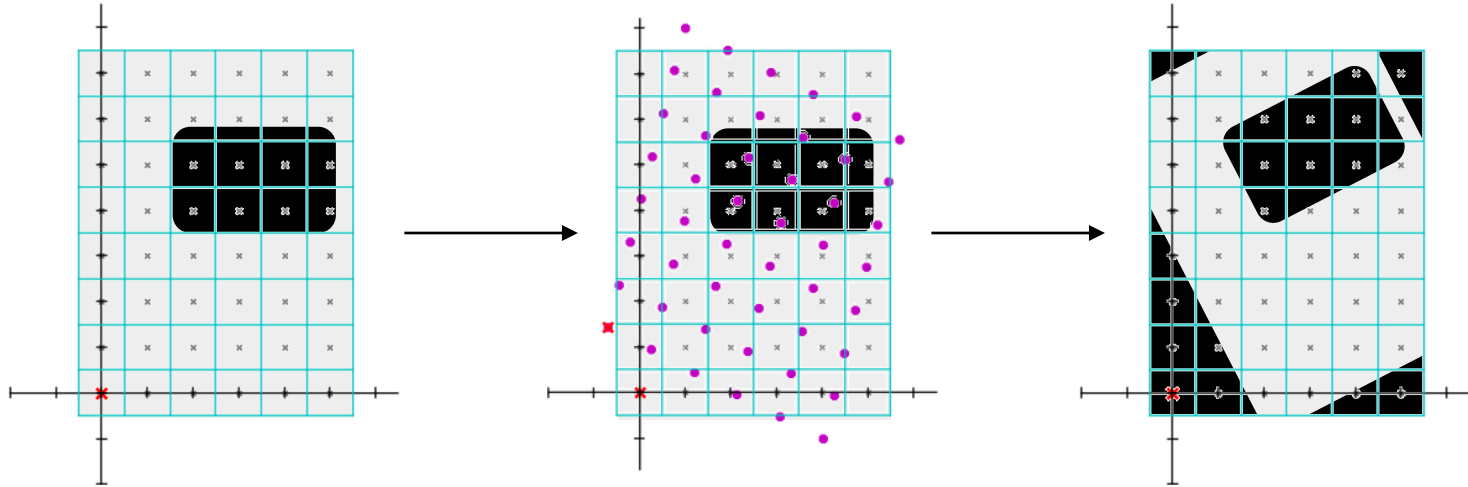
Backward mapping (Euler approach)

- Consider a grid point $\mathbf{y} \in \Omega_F$ in image $M \circ \phi$ and a spatial transformation $\phi : \Omega_F \rightarrow \Omega_M$.
- The intensity value at position $\mathbf{y}$ is set to the intensity value $M(\phi(\mathbf{y}))$ from position $\phi(\mathbf{y})$.
- Interpolation to get intensity value at non-grid points.
- The inverse transformation need to exist for backward mapping! $\phi = \psi^{-1}$



Backward mapping (Euler approach)

A rotation of 20° (of the domain Ω_M of the moving image) results in a -20° rotation of the moving image after resampling (interpolation to grid points)!



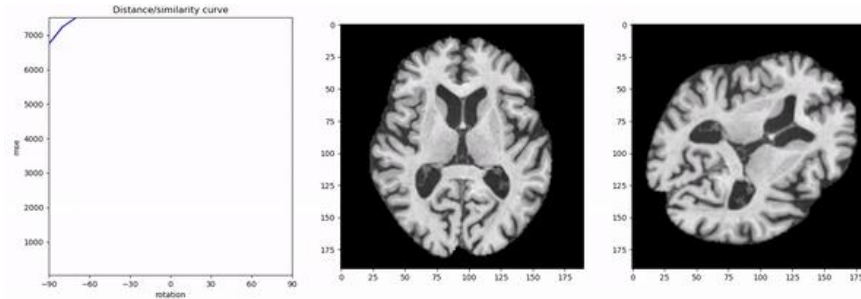
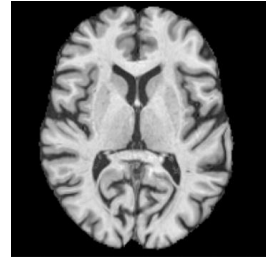
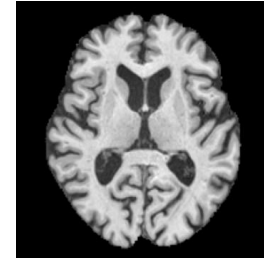
Objective function and Optimization

Mono-modal image similarity

- Objective function $\mathcal{J}(\phi, F, M) = \mathcal{D}(F, M \circ \phi) + \alpha \mathcal{R}(\phi)$
- How can we compute the similarity between two images of the same modality?
- Sum of Squared Differences (SSD)**
 - Assumes identity relationship between intensities

$$SSD(F, M, \phi) = \frac{1}{2} \int_{\Omega_F} (F(\mathbf{x}) - M(\phi(\mathbf{x})))^2 d\mathbf{x}$$

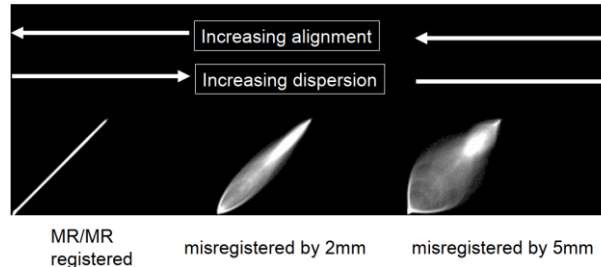
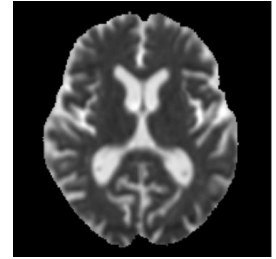
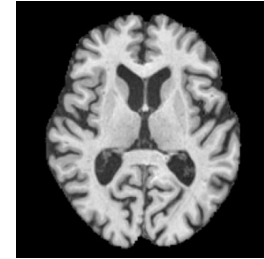
- Normalized Cross Correlation (NCC)**
 - Assumes linear relationship between intensities



Distance curve obtained with SSD

Multi-modal image similarity

- Objective function $\mathcal{J}(\phi, F, M) = \mathcal{D}(F, M \circ \phi) + \alpha \mathcal{R}(\phi)$
- How can we compute the similarity between two images of different modalities?
- Normalized Gradient Fields (NGF)** Haber and Modersitzki, 2007.
 - Assumes intensity changes at the same locations
- Modality Independent Neighborhood Descriptor (MIND)**
 - Exploits self-similarity Heinrich, et al., 2012.
- (Normalized) Mutual Information (NCC)** Viola and Wells, 1997.
Studholme, et al., 1999.
 - Describes how well one image is explained by the other



Joint histograms

Regularization term

- Objective function $\mathcal{J}(\phi, F, M) = \mathcal{D}(F, M \circ \phi) + \alpha \mathcal{R}(\phi)$
- Well-posed problem:
 - a solution exists
 - the solution is unique
 - the solution's behaviour changes continuously with the initial conditions
- For non-parametric image registration problem is ill-posed
 - without further constraints, there are infinitely many solutions
 - the solutions might not be continuous, diffeomorphic, ...
- Regularization term to convert problem into a well-posed problem
- In parametric registration: the search space is restricted by the transformation model
- In non-parametric registration: the problem itself is constraint

Regularization term

- Which regularization term to use?
What kind of deformations are plausible and realistic for the application?

- Smoothness**

Diffusion regularization (1st derivatives)

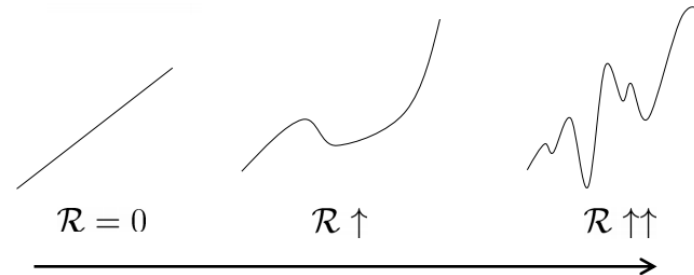
Modersitzki, 2004.

$$\mathcal{R}(\phi) = \frac{1}{2} \int_{\Omega_F} \sum_{j,k}^d \left(\frac{\partial u_k}{\partial x_j} \right)^2$$

Bending energy (2nd derivatives)

Rueckert, et al., 1999.

$$\mathcal{R}(\phi) = \frac{1}{2} \int_{\Omega_F} \sum_{i,j,k}^d \left(\frac{\partial^2 u_k}{\partial x_j \partial x_i} \right)^2$$



Staring, Tutorial@MICCAI 2010.

Regularization term

- Which regularization term to use?
What kind of deformations are plausible and realistic for the application? Diffeomorphisms?

- Smoothness

Diffusion regularization (1st derivatives)

Modersitzki, 2004.

Bending energy (2nd derivatives)

Rueckert, et al., 1999.

$$\nabla \phi(x) = \left(\frac{\partial u_i}{\partial x_j}(\mathbf{x}) \right)_{i,j=1}^{d,d}$$

- Jacobian**

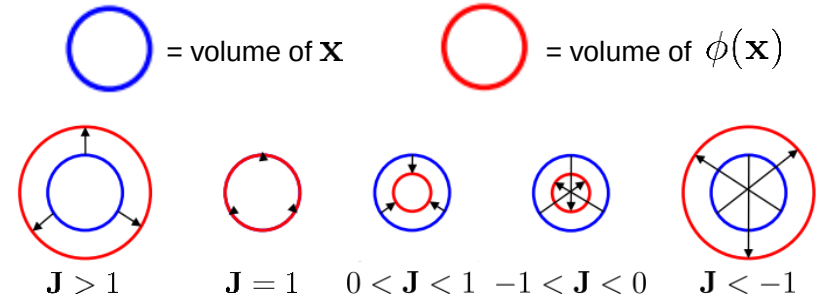
$$\mathbf{J}\phi(\mathbf{x}) = \det \nabla \phi(\mathbf{x})$$

- Other

Elastic regularization

Curvature regularization

Fischer, et al., 2003.



Staring, Tutorial@MICCAI 2010.

$$\mathcal{R}(\phi) = \sum_{\Omega_F} \log \mathbf{J}\phi(\mathbf{x})$$

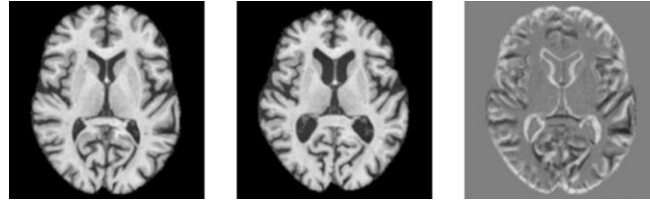
(Incompressibility)

Rohlfing, et al. 2003

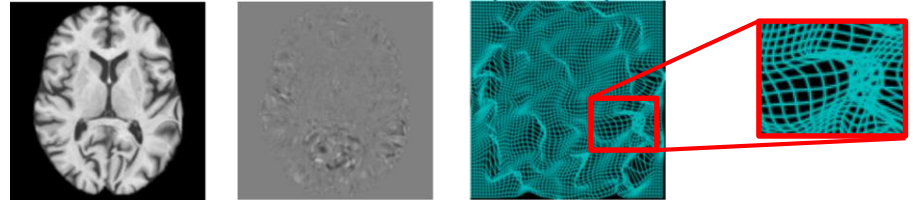
Regularization term: Example

Objective function

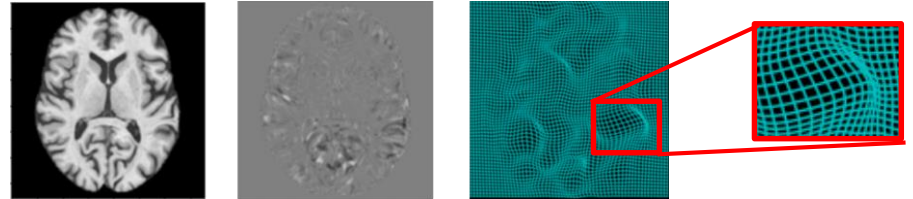
$$\mathcal{J}(\phi, F, M) = \mathcal{D}(F, M \circ \phi) + \alpha \mathcal{R}(\phi)$$



No regularization ($\alpha = 0$)



Regularization ($\alpha > 0$)



Optimization strategies

- Energy minimization

$$\mathcal{J}(\phi, F, M) = \mathcal{D}(F, M \circ \phi) + \alpha \mathcal{R}(\phi)$$

$$\phi^* = \underset{\phi}{\operatorname{argmin}} \mathcal{J}(\phi, F, M)$$

- Discrete optimization: search space is discrete domain (e.g., finite or integers)
- Continuous optimization: search space is continuous (e.g., real numbers)

- Discretization of images, transformations, objective function

- Discrete SSD:
$$SSD(F, M, \phi) \approx \frac{1}{2} \sum_{x \in \Omega_F} (F(\mathbf{x}) - M(\phi(\mathbf{x})))^2$$

- Derivatives:
$$\nabla_{\phi} \mathcal{D}, \nabla_{\phi} \mathcal{R} \qquad \nabla_{\phi} SSD(F, M, \phi) = \sum_{x \in \Omega_F} (F(\mathbf{x}) - M(\phi(\mathbf{x}))) \nabla_{\phi} M(\phi(\mathbf{x}))$$

$$\nabla_{\phi} M(\phi(\mathbf{x}))$$

Optimization strategies

- **Global minimizer:**

$\mathbf{x}^* \in \mathbb{R}^n$ is a global minimizer of function f , if

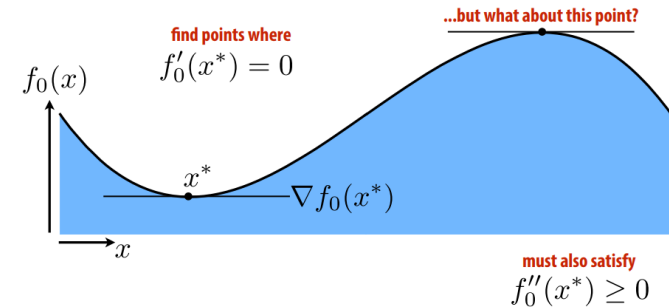
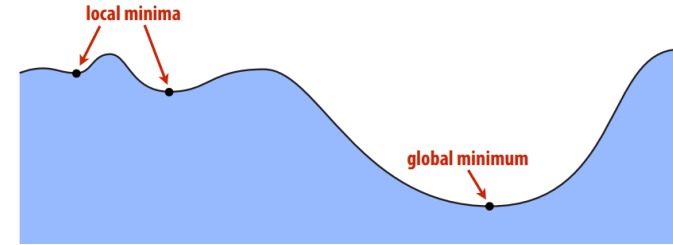
$$f(\mathbf{x}^*) \leq f(\mathbf{x}), \forall \mathbf{x}$$

- **Local minimizer:**

$\mathbf{x}^* \in \mathbb{R}^n$ is a local minimizer of function f , if there is a neighbourhood $\mathcal{N}$ around $\mathbf{x}^*$, such that

$$f(\mathbf{x}^*) \leq f(\mathbf{x}), \forall \mathbf{x} \in \mathcal{N}$$

- We want the global minimizer!
- Note that we optimize over the transformation space!
 n is number of transformation parameters, or number of pixel in dense displacement field.
- Gradient-based approaches



Line search approaches

- Line search approaches:

Initialization $\mathbf{x}_0 \in \mathbb{R}^n$

Search direction $\mathbf{s}_k \in \mathbb{R}^n$

Step length $\alpha_k \in \mathbb{R}^+$

- Gradient descent

Update rule:

$$\mathbf{x}_{k+1} = \mathbf{x}_k + \alpha_k \mathbf{s}_k$$

$$\mathbf{s}_k = -\nabla f(\mathbf{x})$$

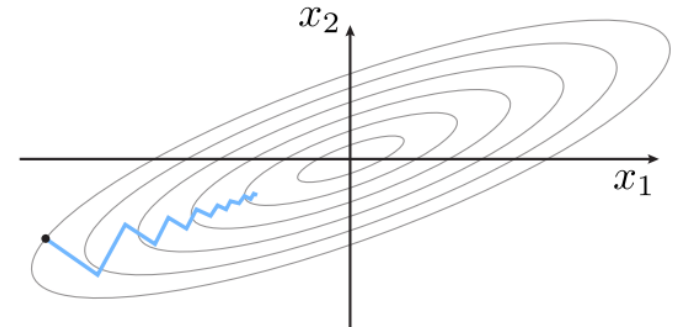
- Higher order derivatives:

Newton

Quasi-Newton

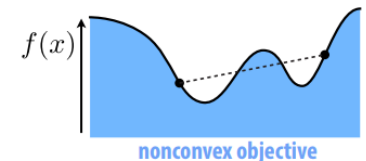
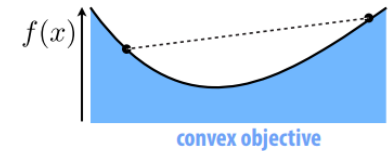
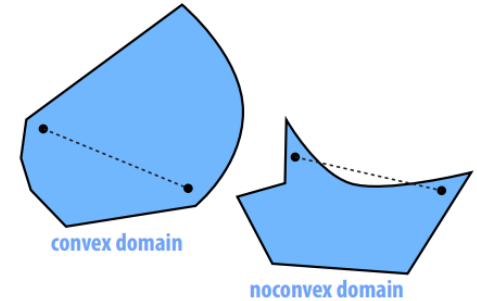
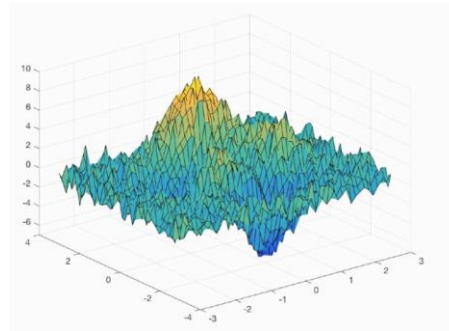
L-BFGS

- Stochastic gradient descent: replaces the gradient by an estimate calculated from a randomly selected subset of the data Klein, et al. *Adaptive stochastic gradient descent optimisation for image registration*. IJCV, 2009.
- Adam is a modified SGD!



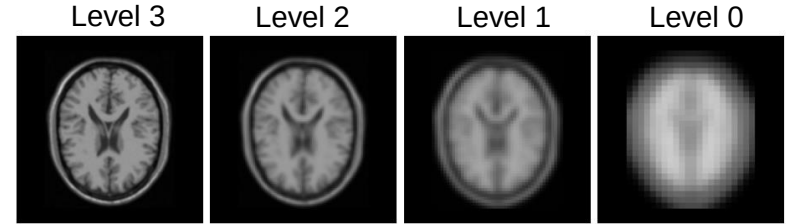
Gradient-based optimization

- Gradient-based approaches work well for convex optimization problems
- Convexity:
 $K \subset \mathbb{R}^n$, K is convex, if
$$(1 - \lambda)x + \lambda y \in K \quad \forall x, y \in K, \lambda \in [0, 1]$$
- The registration problem is typically highly non-convex!
- It needs a good initial transformation for convergence.



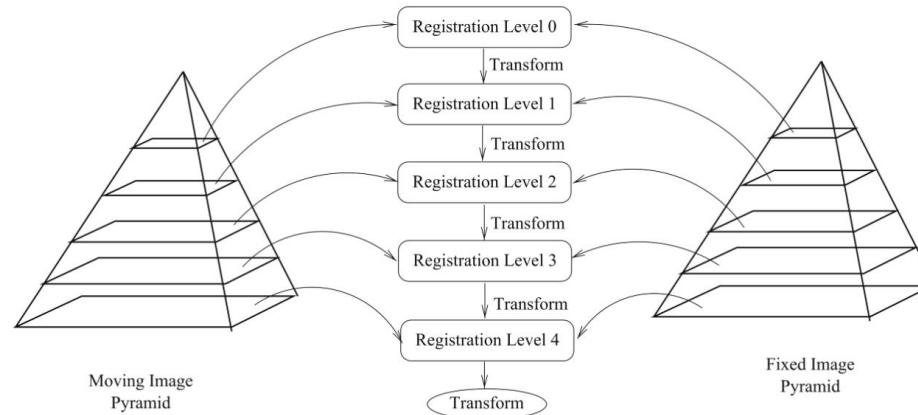
Multi-resolution strategies

- For efficiency
- Better initialization at subsequent levels
- Avoid local minima
- Increase the speed
- Increase robustness



- Increases the pixel size and requires intensity interpolation

→ downsampling →



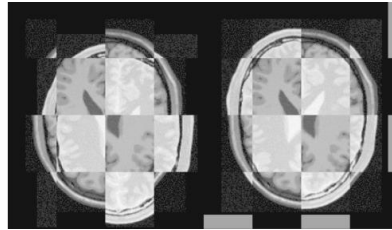
Evaluation

Evaluation of image registration

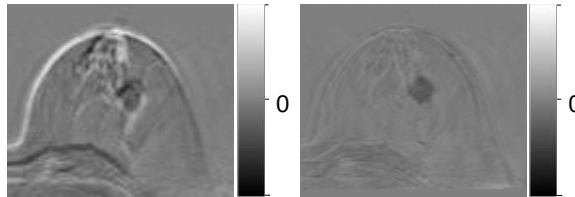
Two types of evaluation:

Qualitative and **Quantitative** evaluation.

Qualitative



Checkerboard before and after



Difference before and after

Quantitative

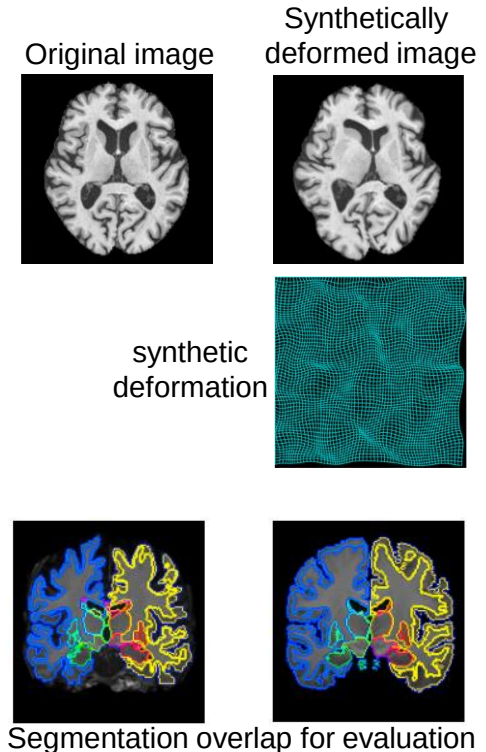
What is a correct registration?

How to define a ground truth?

Quantitative evaluation

Requirement of **ground truth** or **surrogate measures**

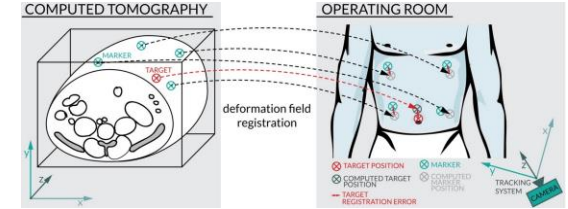
- **Target registration error (TRE):** True error, often unknown
- **Ground truth:** Exact definition of correct transformation
 - Artificial transformations applied to the images
 - Simulations or phantoms
 - hard to define realistically deformed images
 - artificially deformed images are simplified representations of the true registration problem
- **Surrogate measures:** Approximation to correct transformation
 - Segmentation label overlap
 - Landmark error



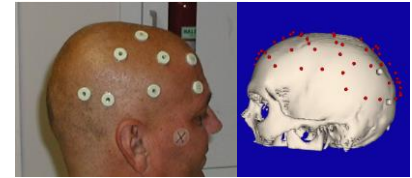
Quantitative evaluation measures

Corresponding points / anatomical landmarks

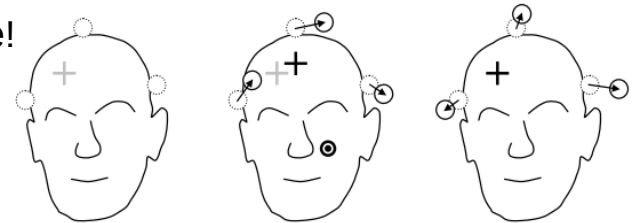
- Fine for linear/global registration, but not for local
- Be careful which landmarks to evaluation on!
- **Fiducial Points** – used to estimate the transformation
- **Target (Fiducial) Points** – independent of transformation estimation
- Fiducial Registration Error and Target Registration Error do **NOT** correlate.
- Fiducial Registration Error is **NOT** a reliable evaluation measure!



Sylwester and Spinczyk, 2018.



Evaluation needs to be independent of registration features or cost function!



Quantitative evaluation measures

Corresponding points / anatomical landmarks
(linear registration)

Similarity measures (not used for registration!)

**Evaluation needs to be independent of registration
features or cost function!**

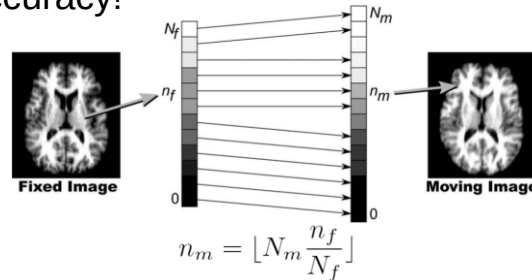
Quantitative evaluation measures

Corresponding points / anatomical landmarks
(linear registration)

Similarity measures (not used for registration!)

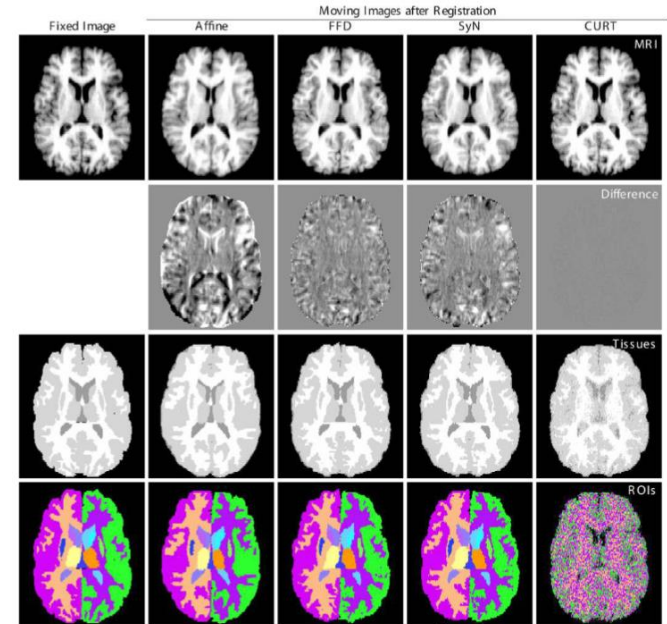
Segmentation label overlap

- Segmentation overlap (of large structures) is not a reliable surrogate for registration accuracy!



Evaluation needs to be independent of registration features or cost function!

CURT (completely useless registration tool)



Quantitative evaluation measures

Corresponding points / anatomical landmarks
(linear registration)

Similarity measures (not used for registration!)

Segmentation label overlap

Quality of deformation field (e.g., consistency, smoothness, volume loss, topology, ...)

Simulations

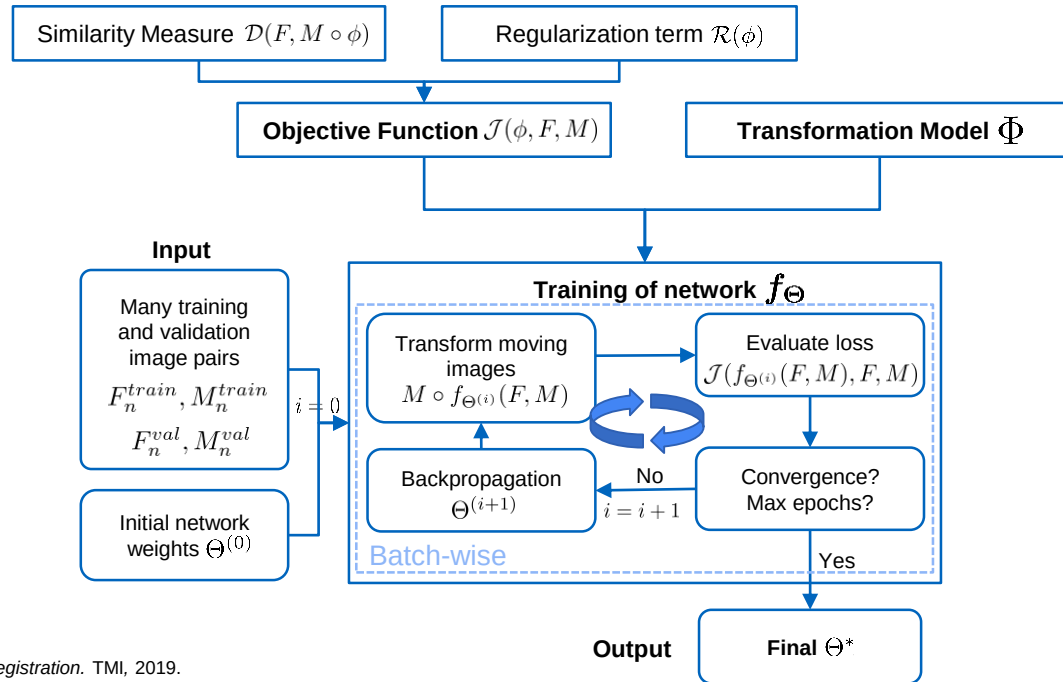
- Artificial transformations
- Physical models/ phantoms

**Evaluation needs to be independent of registration
features or cost function!**

Learning-based Image Registration

Learning-based image registration algorithm

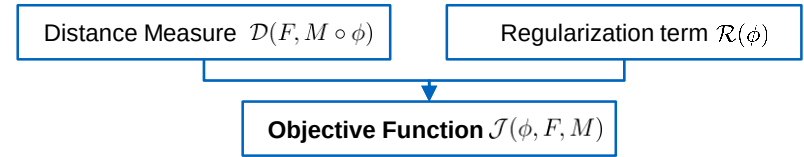
- Learn the **spatial transformation**
- Exploit information of many image pairs
- Optimization of many training samples instead of over one image pair
- CNN f_{Θ} parametrised by Θ to encode transformation
- Differentiable image sampling
Jaderberg, et al., 2015.
- **Supervised**
- **Unsupervised**
- **Weakly supervised**



Balakrishnan, et al., *VoxelMorph: a learning framework for deformable medical image registration*. TMI, 2019.
De Vos, et al., *A deep learning framework for unsupervised affine and deformable image registration*. MedIA, 2019.

Other learning-based approaches

- **Multi-resolution approaches**
- **Iterative deep approaches**
- Learn the **image similarity / distance measure** or **regularization**

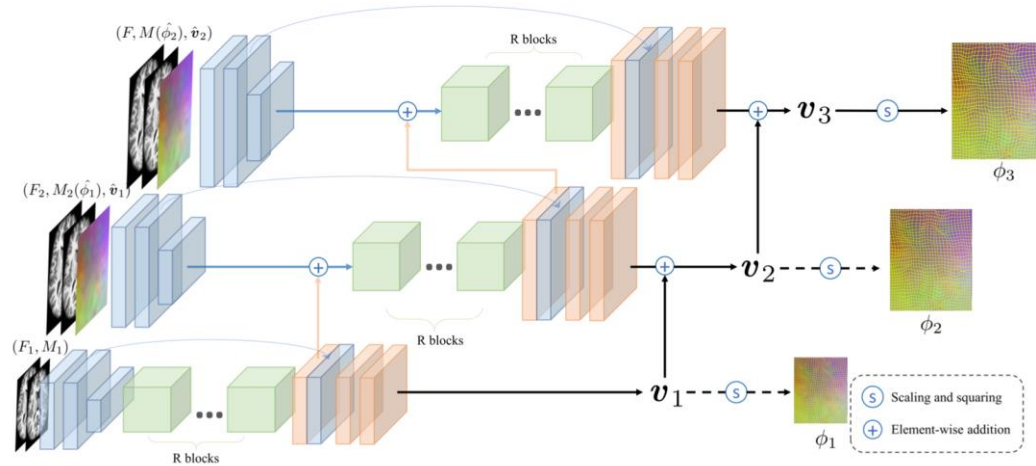


Deep image similarity:

- CNN f_{Θ} parametrised by Θ to encode distance between images
- Pairwise iterative optimization to find ϕ^*

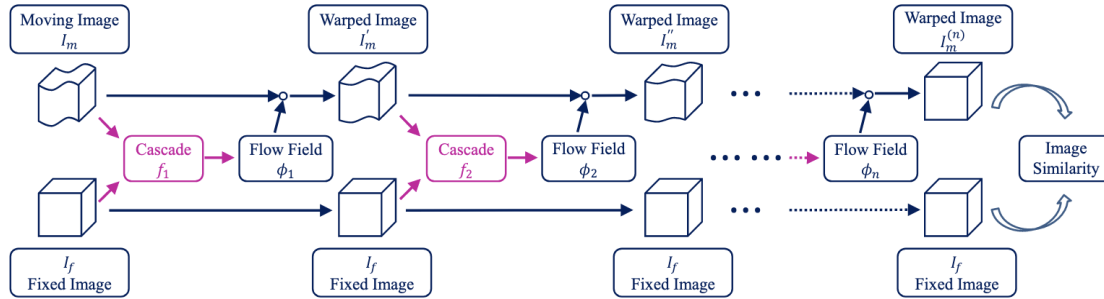
$$\phi^* = \arg \min_{\phi} f_{\Theta^*}(F, M \circ \phi) + \alpha \mathcal{R}(\phi)$$

Multi-step and multi-resolution networks

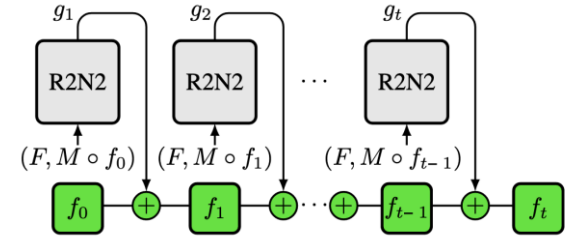


- Mok, T.C.W., Chung, A.C.S.: Large Deformation Diffeomorphic Image Registration with Laplacian Pyramid Networks (MICCAI 2020)
- de Vos, B.D., Berendsen, F.F., Viergever, M.A., Sokooti, H., Staring, M., Išgum, I.: A deep learning framework for unsupervised affine and deformable image registration. (Med. Image Anal. 2019)
- Hering, A., van Ginneken, B., Heldmann, S.: mVIRNET: Multilevel Variational Image Registration Network (MICCAI 2019)

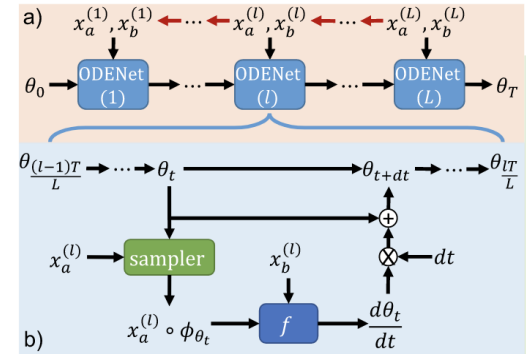
Multi-step and multi-resolution networks



Zhao et al. CVPR 2019



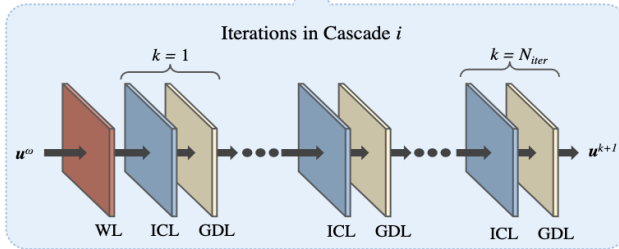
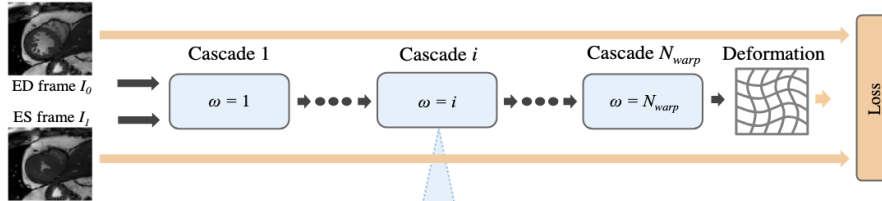
Sandkühler et al. NeurIPS 2019



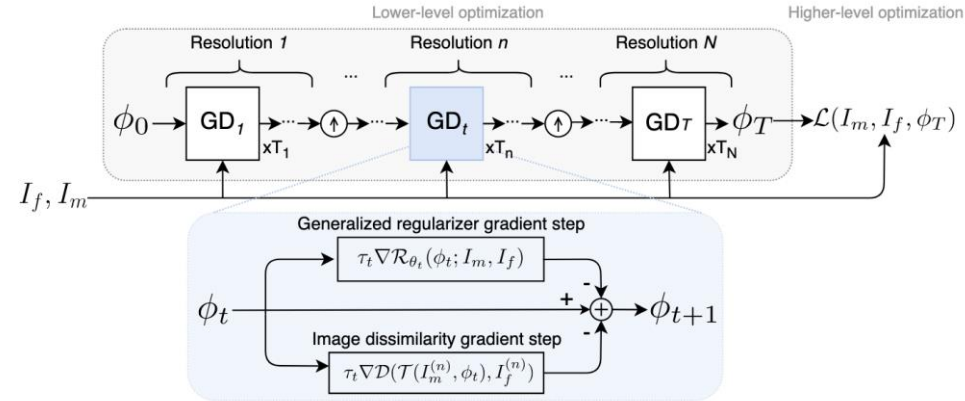
Xu et al. MICCAI 2021

- Zhao, S., Dong, Y., Chang, E.I.-C., Xu, Y.: Recursive Cascaded Networks for Unsupervised Medical Image Registration, (CVPR 2019)
- Sandkühler, R., Andermatt, S., Bauman, G., Nyilas, S., Jud, C., Cattin, P.C.: Recurrent Registration Neural Networks for Deformable Image Registration (NeurIPS 2019)
- Xu, J., Chen, E.Z., Chen, X., Chen, T., Sun, S.: Multi-scale Neural ODEs for 3D Medical Image Registration (MICCAI 2021)

Unrolled optimization



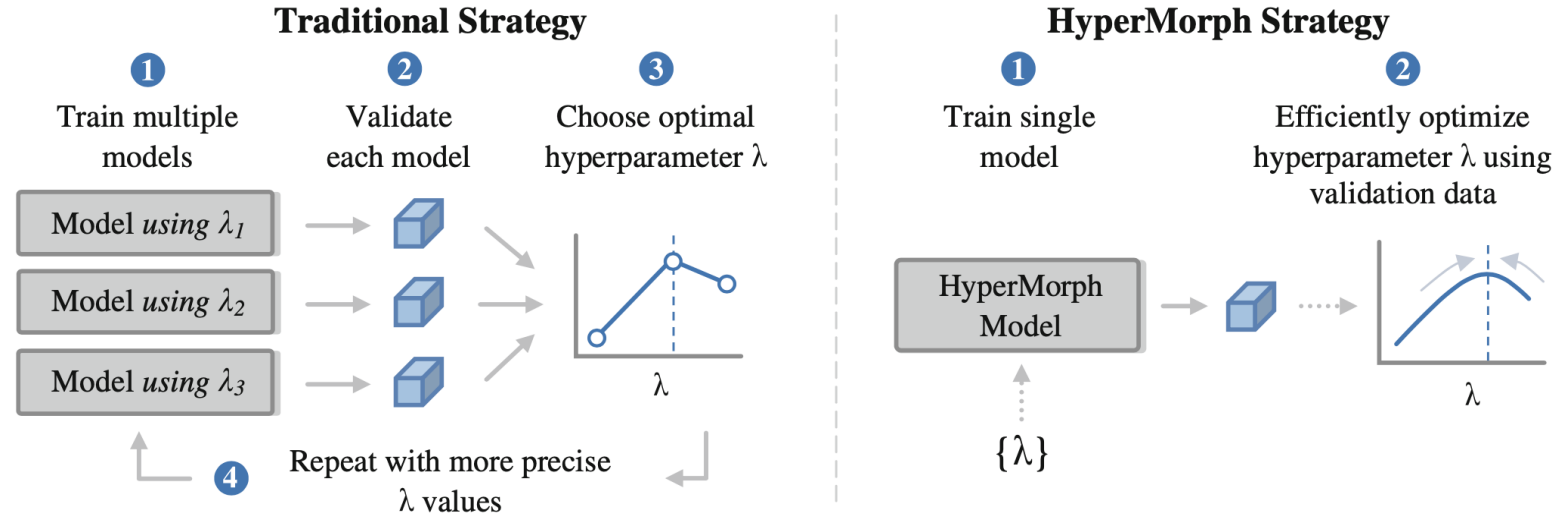
Jia et al. IEEE TMI 2021



Qiu et al. MICCAI 2022

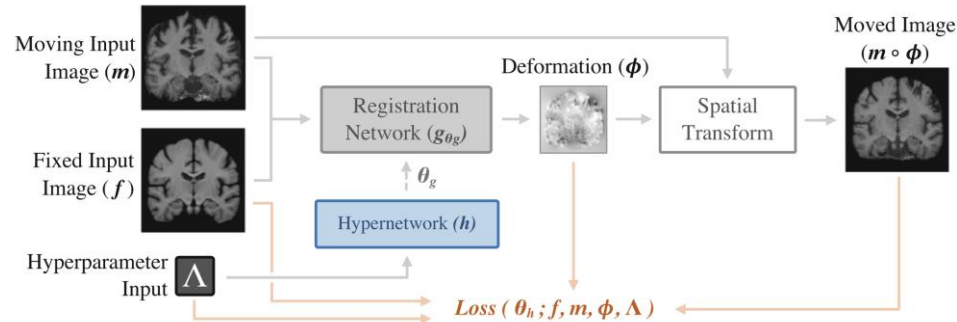
- Qiu, H., Hammernik, K., Qin, C., Chen, C., Rueckert, D.: Embedding Gradient-based Optimization in Image Registration Networks (MICCAI 2022)
- Jia, X., Thorley, A., Chen, W., Qiu, H., Shen, L., Styles, I.B., Chang, H.J., Leonardis, A., De Marvao, A., O'Regan, D.P., Rueckert, D., Duan, J.: Learning a model-driven variational network for deformable image registration. (IEEE Trans. Med. Imaging 2021)

Learning regularization: hyperparameter

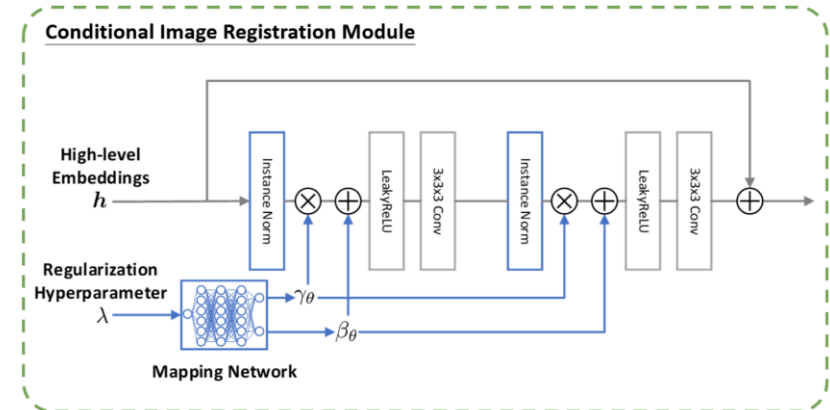


- Hoopes, A., Hoffmann, M., Fischl, B., Gutttag, J., Dalca, A.V.: HyperMorph: Amortized Hyperparameter Learning for Image Registration, (IPMI 2021)
- Mok, T.C.W., Chung, A.C.S.: Conditional Deformable Image Registration with Convolutional Neural Network, (MICCAI 2021).

Learning regularization: hyperparameter



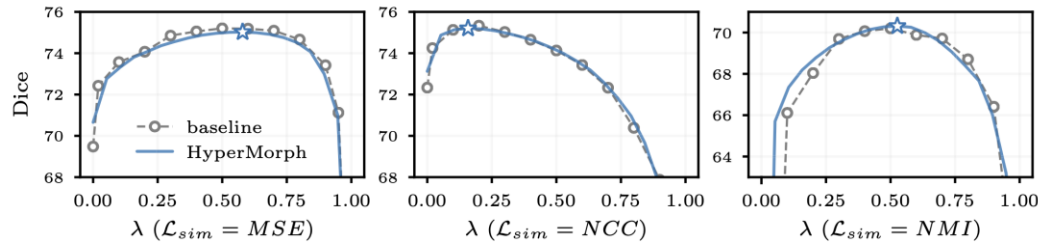
Hoopes et al. IPMI 2021



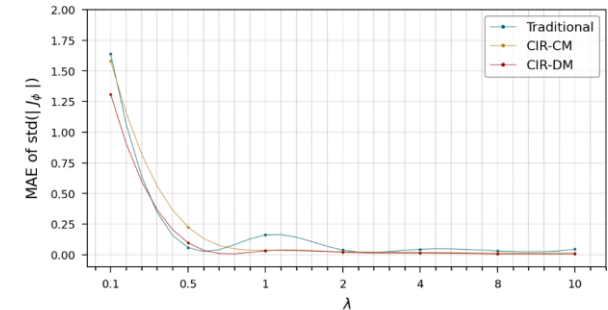
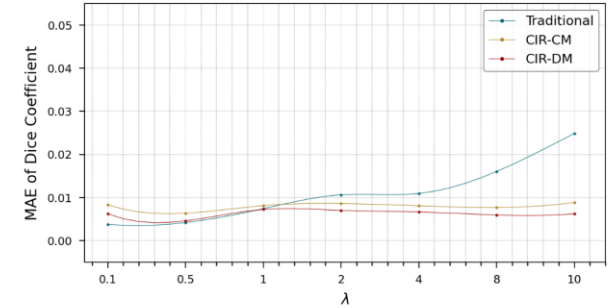
Mok et al. MICCAI 2021

- Hoopes, A., Hoffmann, M., Fischl, B., Gutttag, J., Dalca, A.V.: HyperMorph: Amortized Hyperparameter Learning for Image Registration, (IPMI 2021)
- Mok, T.C.W., Chung, A.C.S.: Conditional Deformable Image Registration with Convolutional Neural Network, (MICCAI 2021).

Learning regularization: hyperparameter



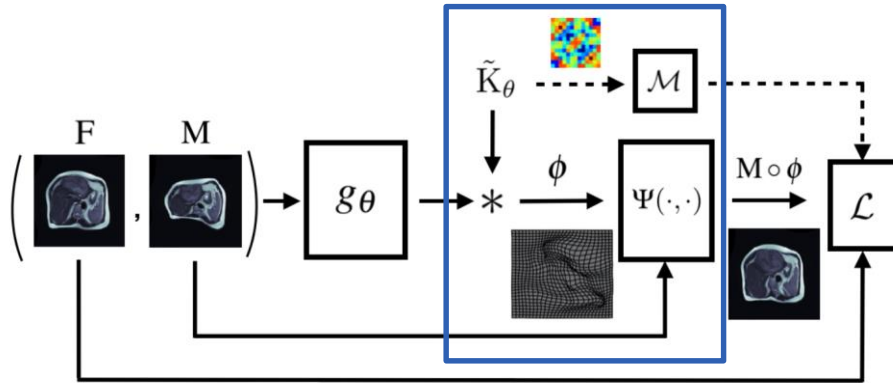
Hoopes et al. IPMI 2021



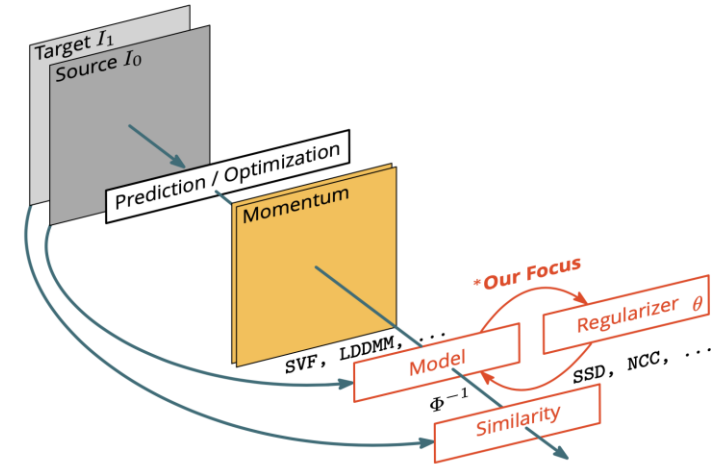
Mok et al. MICCAI 2021

- Hoopes, A., Hoffmann, M., Fischl, B., Gutttag, J., Dalca, A.V.: HyperMorph: Amortized Hyperparameter Learning for Image Registration, (IPMI 2021)
- Mok, T.C.W., Chung, A.C.S.: Conditional Deformable Image Registration with Convolutional Neural Network, (MICCAI 2021).

Learning regularization



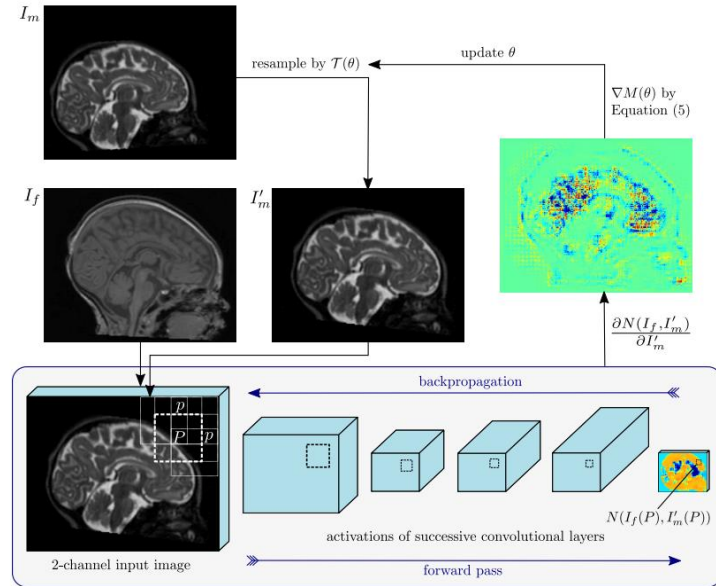
AI Safadi et al. CVPR 2021



Niethamer et al. CVPR 2019

- AI Safadi, E., Song, X.: Learning-Based Image Registration With Meta-Regularization. (CVPR 2021)
- Niethammer, M., Kwitt, R., Vialard, F.-X.: Metric Learning for Image Registration. (CVPR 2019)
- Shen, Z., Vialard, F.-X., Niethammer, M.: Region-specific Diffeomorphic Metric Mapping. (NeurIPS 2019)

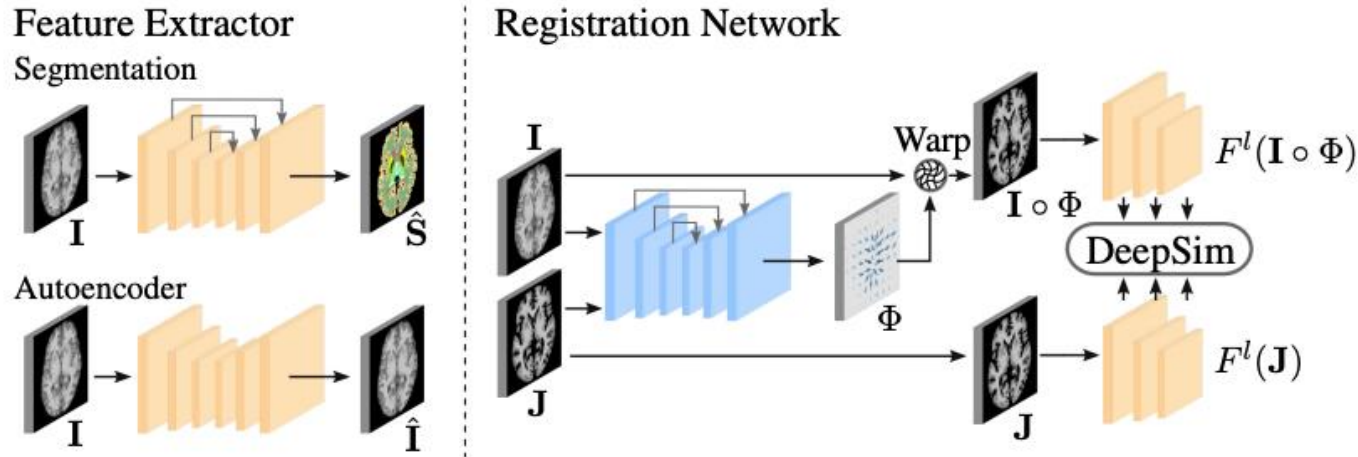
A Deep Metric for Multimodal Registration



- Train a classifier to classify patches as match or mismatch
- Positive and negative pairs generated from pre-aligned images
- Output of the trained classifier as the similarity metric

Simonovsky, M., Gutiérrez-Becker, B., Mateus, D., Navab, N., Komodakis, N.: A deep metric for multimodal registration. In: International Conference on Medical Image Computing and Computer-Assisted Intervention, pp. 10–18. Springer, Berlin (2016)

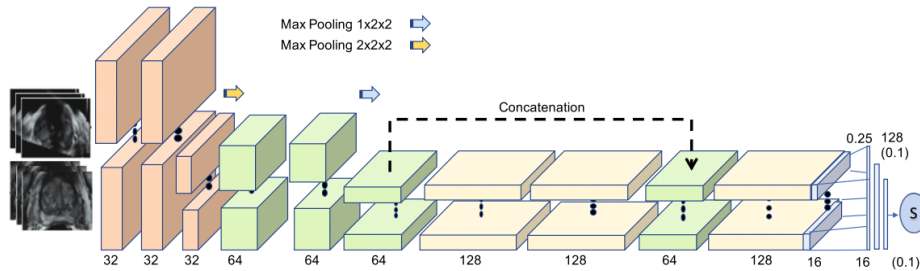
Semantic similarity metrics for learned image registration



A data-driven similarity metric based on the alignment of semantic features.

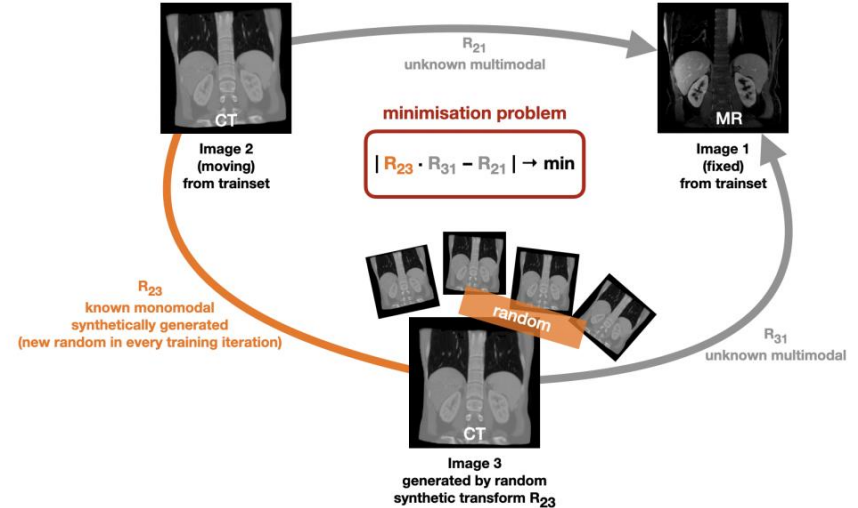
Czolbe, Steffen, Oswin Krause, Aasa Feragen. 'DeepSim: Semantic similarity metrics for learned image registration'. In: *Medical Imaging with Deep Learning*. PMLR, 2021.

Learn a similarity metric for rigid registration



Learn the target registration error (TRE) between 3D MR and TRUS images. (supervised) [1]

[1] Haskins, Grant, Jochen Kruecker, Uwe Kruger, Sheng Xu, Peter A. Pinto, Bradford J. Wood and Pingkun Yan. "Learning deep similarity metric for 3D MR-TRUS image registration." *International Journal of Computer Assisted Radiology and Surgery* 14 (2018): 417-425



Geometric self-supervision – minimise the discrepancy created through a cycle of transformations [2]

[2] Siebert, Hanna, Lasse Hansen and Mattias P. Heinrich. "Learning a Metric for Multimodal Medical Image Registration without Supervision Based on Cycle Constraints." *Sensors (Basel, Switzerland)* 22 (2022).

Summary and conclusions

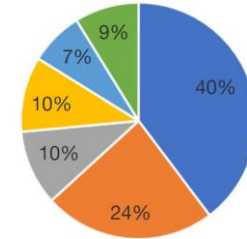
- (Quick) overview of the generic image registration problem.
- Image geometry and image warping
- Main building blocks and types of registrations
- Learning-based approaches for image registration

What are the next challenges?

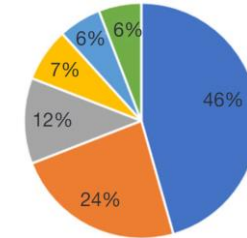
- Clinical images are often ugly
- Domain adaptation / Out-of-distribution data
- Uncertainty estimation

What we have not talked about:

- Discrete optimization techniques, Diffeomorphic registration, ...
- Use of graphs, GANs, ...



■ Brain ■ Lung ■ Prostate ■ Abdominopelvic ■ Liver ■ Others



■ MR-MR ■ CT-CT ■ MR-CT ■ MR-US ■ CT-CBCT ■ Others

Xiao, et al., 2021.

Thanks!